Humble Lamppost

a kick start for a smart city's journey



Management Framework

Packaging smart cities solutions

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Foreword

This management framework is intended for SCC01 Lighthouse cities, or indeed any other cities that may wish to implement smart lampposts. This guidance document was prepared through the collective knowledge from Sharing Cities and other lighthouse programme partners. More information about Sharing Cities and the lighthouse programme can be found here: http://www.sharingcities.eu/ and https://eu-smartcities.eu/group/454/description

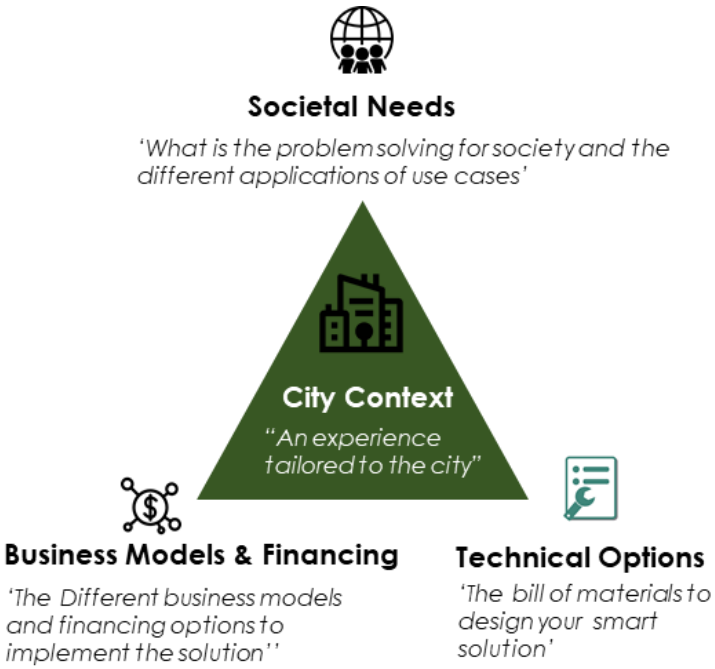
What is a 'Management Framework?

A management framework is primarily written for cities. It seeks to reduce the effort, speed the process, strengthen quality and confidence in outputs, align across disciplines, and generally ready a city to engage the market to acquire a solution. It does so through offering a step-wise approach, suggesting pragmatic tools that can be applied, and identifying the key issues that need to be addressed in moving ideas to implementable projects.

This is one of a series of 'Packaging' documents focused on the initial task of supporting stakeholder engagement. The **Leadership Guide** and **Smart Booklet** provide complementary information for their intended audiences. The **Smart Leaflet** offers a model document to engage society.

What is 'Packaging'?

'Packaging' supports cities to help reduce the effort, speed the process, deliver greater consistency, and add value through implementation. The intention of packaging is to provide a trusted familiar source of structured guidance. It will enable scale up smart city solutions and reduce risk by basing their decisions on tested methods. Packaging comprises a portfolio of templates and documents that offer guidance and support from idea to operation of smart measures, based on the experience of the lighthouse programmes. For each measure 'packaging' addresses (i) societal needs, (ii) technical options, (iii) business models and financing. Importantly, it also offers insight in the specific context of the cities, highlighting similarities and differences that can resonate with other cities' contexts to enable appropriate local tailoring of solutions.



Portfolio of packaging documents

- The packaging portfolio intends to deliver a series of quality documents from a consistent framework, written at a length and style to suit the recipient and the right time; recognisable and that builds familiarity and trust in the market.
- The portfolio represents a number of documents covering 4 project stages. The purpose of the Management Framework is to support Stage 1 Audience Engagement, specifically to support the further development of a smart city solution by the nominated lead officer(s) as it moves from idea to commitment, and prior to action.
- ***Directed towards**: Political Leaders; Policy Advisors; City Officers; City Technicians; Industry; Investors; Citizens*

	Documents	Contents	Audience
Stage 1 Audience Engagement	1. Leadership Guide	Short (~6p) 'train read': context; proposition; value; issues; actions	City Leaders
	2. Smart Booklet	10p Measure Definition, Benefits, Impact (What & Why)	City Officers
	3. Management Framework	**Core document (~40p) Societal Needs; Specifications; Business models; city context**	**City Officers**
	4. Smart Leaflet	1p brochure: benefits; key facts; pictures	Citizens & People
Stage 2 Making the Case	1. Business Value Case	Financial and Non-Financial value case; economic modeling where appropriate	City Officers
	2. Business Models	List different business models and financing options	City Officers
	3. Investor Guide	10p guide; Investor framework; value and impact on each measure	Investors
	4. Market Outlook	5 pages guide focusing the city context, market context, EU and World context	City Officers
Stage 3 Implementation	1. Supplier Listing	Market & supplier information structured and detailed with sub-components	City Technicians
	2. Procurement Templates	Procurement templates for use by buyer, however known by suppliers	City Technicians Industry
	3. Technical Specifications	50+p design specifications on each feature; may well be multiple docs/references	City Technicians
	4. Standards Listing	Searchable listing (e.g. by jurisdiction)	Industry
Stage 4 Capturing Value	1. Indicator Listing	Listing indications (f and non fi) and issues; data availability/quality	Cities/ Investors
	2. Replication Scale Guide	Short guide 10 pages including bench learning, collaboration (what/how/why),	Cities
	3. Demand Aggregation	~10-20p addressing city context, governance, BM&F, Procurement and Decisions	Cities
	4. Market Case Studies	10-15P Short structured document capturing the different market case studies	Cities

Summary

This document is written for cities. To provide a practical guide through the step-wise process of evaluating and shaping a smart lamppost solution then implementation.

Smart lampposts offer a city the opportunity to exploit its existing lighting assets; to combine the upgrade of them to low energy luminaires to make very significant energy and operational savings; and in parallel to add 'smart' features (WiFi, environmental monitoring, Electric Vehicle charging and the like).

Doing so also opens up the opportunity to consider additional value-add through collaborating with other cities around a common design, learning, approach and specification to benefit further through collaboration in the market.

The smart lamppost is also one of the more obvious smart city 'quick wins' that a city can implement that has the advantage of supporting them on their more strategic smart city roadmap.

This document is an open deliverable that works alongside a number of other referenced documents (e.g. specifications; supplier listings, etc) and is intended for broad use by European cities. In time, it will be offered to the international Standards Development Organisations for adoption/adaption and publication.

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1 Introduction

This section sets out the readership and purpose of the document; summarises what we mean by smart lampposts; discusses activities at a European scale; tables the principles that underpin this component-based approach; and introduces the (current) dozen or so elements of the "Smart Lamppost Toolkit".

1.1 Purpose

This Management Framework is written by and for the Sharing Cities cities; and any other city that may wish to upgrade their lighting stock and install smart services. The target reader may have a variety of functional backgrounds, however will likely be the person(s) who holds, or may hold responsibility to implement smart lampposts. As this is written for use beyond the Sharing Cities programme we limit the reference to that programme. (More on Sharing Cities can however be found at www.sharingcitites.eu).

The purpose of this document is to provide a practical 'how to' guide – a 'toolkit' – for cities that seek to implement smart lampposts. The document will:

- help **ready cities** for swifter action
- provide greater **confidence** within cities
- provide a **common basis** that supports collaboration between cities
- potentially **stimulate demand aggregation**, which can lead to better market value (notably for smaller cities that cannot bring scale to the market)
- **exploit what already exists** – adopt or adapt; only create anew where necessary

1.2 Definition of 'Humble' Lamppost "The What"

The 'Humble Lamppost' comes from the Humble Lamppost initiative launched under the European Innovation Partnership for Smart Cities and Communities (EIP-SCC) to raise awareness about exploiting lighting assets. 'Humble' reflects an asset that is widespread in every city yet rarely considered of much value – too often neglected.

Pragmatically, we assume 'smart' lamppost systems are those that:

- retain, serve and indeed enhance the principal compliant obligations of street lighting (way-finding, public safety)
- use low-emission luminaires
- are connected as a network, enabling system-wide controllability (e.g. central mgmt. system), and integration of smart services
- include within the network a number of additional smart services (sensors and the like) that extend services beyond just the provision of light
- arguably, have 24x7 power to enable continuous smart services

Thus, a "smart lamppost", with its various services, will most likely be mixed within a larger portfolio of existing 'traditional' lampposts (that have been upgraded to light emitting diode (LED) lighting.

1.3 Value Proposition "The Why": The Smart Lamppost Opportunity at a European Scale

As context for this document, we provide a summary of the opportunity for smart lamppost implementation across Europe.

The ***EIP-SCC (European Innovation Partnership for Smart Cities & Communities)*** seeks to bring together cities, industry, investors and others to accelerate uptake, at scale, of smart city solutions through active managed collaboration. 5,000 organisations made commitments to the cause at launch in 2013. There are around 20 significant initiatives within the 6 action clusters that frame the partnership.

The 'Humble Lamppost' is one of the more developed initiatives, and seeks to upgrade 10 million street lights across EU cities. The initiative is considered an obvious 'quick win' for smart cities, and the significant goal has attracted the attention of senior politicians, financiers, and other stakeholders. The case for action is strong, as can be seen in figure 1. The logic being that the economic motive to upgrade for energy savings stimulates action to also capture the various other 'smart' service

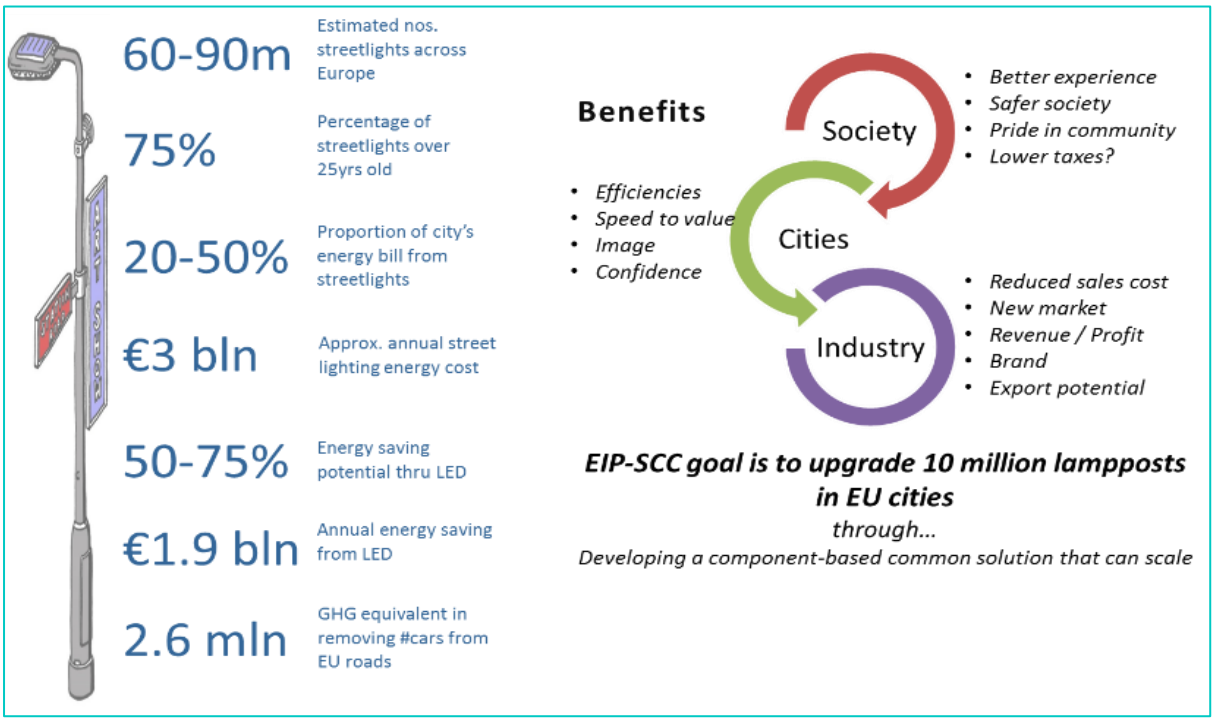


Figure 1 The EIP-SCC "Humble Lamppost" Initiaitve case for action

opportunities that are discussed further in this document. In addition to the EIP-SCC initiative, there are now 14 Horizon 2020 Smart City "Lighthouse" programmes in action across Europe that involve overall ~100 cities. These provide an ideal and incentivised grouping of target cities to support uptake of such an initiative. The percentage of luminaires across EU that have been upgraded to LED is low – single or low double digit percentage. This is however increasing rapidly as the attractive savings / ROI and market confidence in the technical solutions becomes apparent. The supply market is also increasingly active – also on the smart services standpoint – as the potential to multi-purpose lampposts becomes more apparent. This suggests that the time to act is now. And if EU industry does act swiftly (led of course by the actions of cities), there are clear economic opportunities, both within Europe and in the rest of the world.

1.4 Principles that Underpin this Component-Based Approach

Ten principles set the foundation by which cities will maximize value from this approach:

1. **City-needs-led** approach, that ensures a clear demand signal forged by the use cases and user needs of the city
2. **Component-based core design,** that enables common shared solutions to be implemented
3. '**Building blocks**' that support *tailoring* of solutions to city-specific needs
4. **Adaptable to cities** of different size, and stage of development (notably offering value for smaller cities)
5. A '**digital first' approach** that seeks to maximize value from multi-functional technologies and city data
6. **Integrated, open solutions** that prevent vendor lock-in, and enable rapid capture as standards
7. **Early engagement in the project value chain,** to capture opportunities for innovation and support acceleration of adoption
8. **Multi-disciplinary perspective** to help cross the various sectors and 'silos' in cities
9. **Deployed at scale**, through demand aggregation and joint acquisition, application of common requirements, or consistent replication activities
10. Incorporation of EU and worldwide **leading practices**

These principles are addressed in the approach that follows.

1.5 Introducing the 'Humble Lamppost' Toolkit

A set of tools such as templates, guidance documents, and methodologies that will provide support for cities to implement the smart lamppost in a number of areas throughout the value chain. The toolkit includes ~20 steps covering five stages of project development. Depending on which stage a city is at, the toolkit will help at any stages in implementing the humble lamppost.

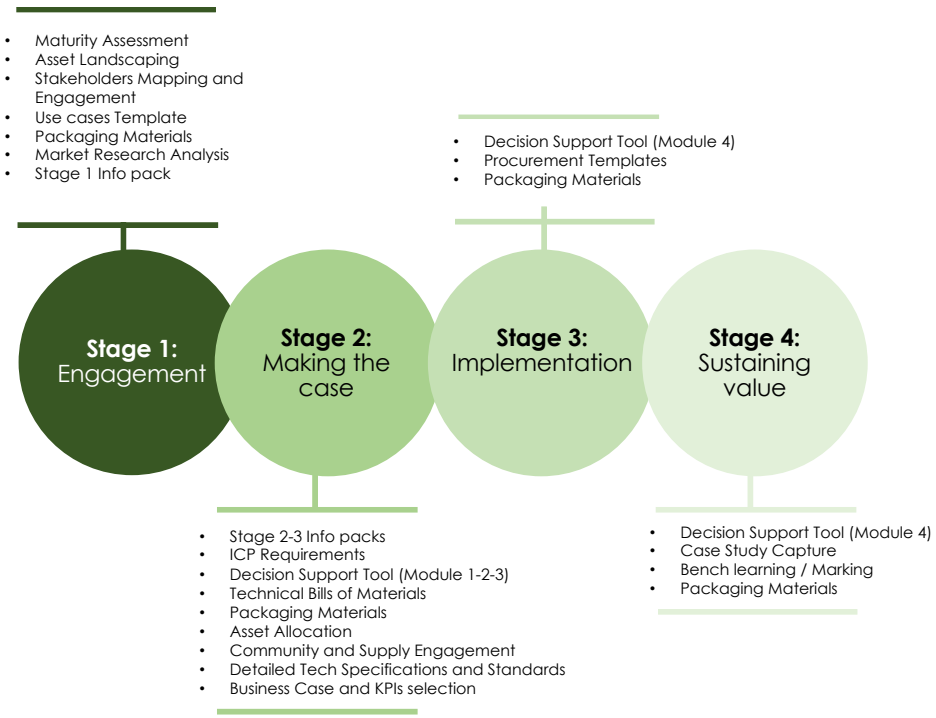


Figure 2 Humble Lamppost Implementation Toolkit

Stage 1 Engagement: Considering current state and what / where the opportunities lie and Engaging the relevant stakeholders and aligning to city priorities

- **Maturity / Development Model** – this is a 5-step common frame of reference that can be used to capture current state, and desired ambition, against a set of interdependent themes.
- **Asset Landscaping** – this is an important step in the process to capture the current (smart) lighting assets. The current process tends to be city specific. There are emerging tools that can help streamline information collection.
- **Use Case Templates** – use cases provide a description in very tangible terms for the value that occurs from implementation of smart services. We make explicit links to *DIN PAS 91437 "imHLa - integrated multi-functional Humble Lamppost"*.
- **Stakeholder Mapping and Management** –the recognition of who is engaged to ensure support at the right point in time is vitally important. There are tools that can help this process.
- **Packaging Engaging Materials** – advising and convincing political and professional city leaders to make decisions to implement smart lighting is important and non-trivial. As is the task of supporting the management of the various sectors (public and private) that will own and manage the resulting services. Materials to support this are in development and the intention is to formalize these through the (BSI) standards process. This will complement the DIN materials.
- **Market Research & Analysis** – the market for the various elements of the smart lamppost differs significantly across Europe, so a local assessment is typically warranted (if one is not available from an analyst or fellow city). For some components of the solution information will be relevant and available that are internationally applicable (e.g. luminaires); however, for other elements (e.g. like lighting columns, or sensor solutions) that may be available locally it is important to assess the local market.
- **Stage 1 Info pack**: A pragmatic tool and template to collect initial asset information to inform plans and ambitions and engage the right people within the city.

Stage 2 Making the case: getting the troops ready before implementation

- **Asset Allocation** provides a method to appropriately allocate use cases, and thus equipment, in the various city district types.
- **Stage 2-3 Info pack**: A more detailed data collection template guiding officers to capture essential information and data to make the business case
- **Detailed Specifications** – many standards exist that provide technical detail for street lighting. And plans may well be in place to update existing or develop more, in addition to the noted DIN and BSI activities. Few/none of the more than 300 potentially relevant standards screened are: international; deal with aspects of finance and performance; or discuss smart services. However, several will be relevant. We seek to identify which documents cities could/should use.

- **Technical "Bill of Materials"** – in order to implement a modern smart lamppost system the city must develop a comprehensive listing (bill of materials) for their intended new system.
- **Business Case & KPIs Selection**– the financial and non-financial value from upgrading to LED and implementing smart services; and the associated uncertainties and risks, require the development of a more comprehensive justification. Tools to support this that provide confidence and consistency are thus essential to help accelerate collaborative implementation. Regarding the KPIs selection a template provides a set of standard indicators to start with.
- **Business Model & Financing** – ownership and operational accountability for lighting and smart services associated with lampposts can involve various actors. Options can be developed; choices must be made around this important theme as it can set boundaries of possibilities into the future. Public budgets are under strain, so in many circumstances' cities may need incentives to move at scale (beyond a solid business case / RoI). Financing sources and options are explored – also in relation to business case and business models.
- **Decision Support Tool (Module 1,2 and 3)**: a comprehensive digital platform to evaluate bankable smart lighting projects through a set of modules. Module 1 supports the use cases selections. Module 2 defines the project boundaries and indicates cost and value. Module 3 suggests tailored business models and financing options.
- **ICP Requirements**: The protocol requirements from the Investor Confidence Project for energy efficiencies projects which provides certifications.
- **Packaging Engaging Materials** – again a range of other supporting guidance materials for key city decisions makers as well as investors e.g. Investor Guide; User Guide DST; …
- **Community and Supply Engagement**: Both groups are vitally important to engaged before implementing Smart Lampposts to ensure adoption from the public and the engage of industries to inform cities on procurement, costs and value. This process will describe the type of the vent to do these sorts of engagement.

Stage 3 Implementation: Technical & Specifications details

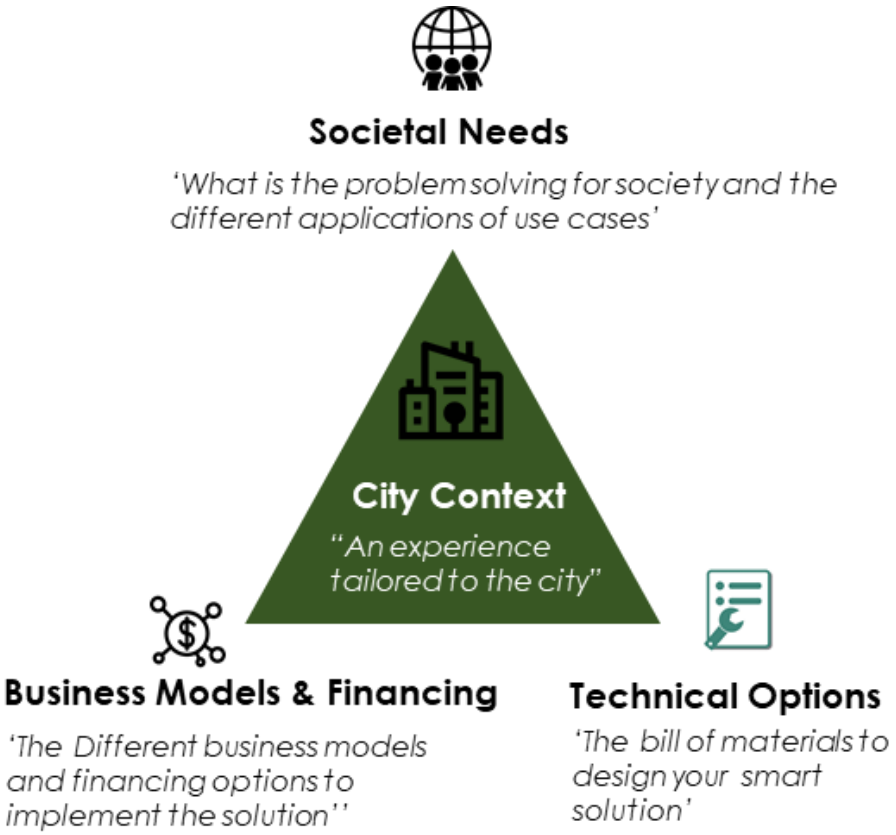
- **Procurement / Commissioning Templates** – speeding and guiding cities through the acquisition process is vitally important. The toolkit captures some of the basic frameworks and content that cities can use to aid this process. Here we recognise that nation states and cities may well have local policies and practices that play a major shaping role in this. And we recognise that the European Commission is seeking to streamline regulation and processes to enable cross-EU procurements to become more commonplace.
- **Packaging Engaging Materials** – again a range of other supporting guidance materials for key city officers and technicians e.g. Supplier Listing
- **Decision Support Tool (Module 4)**: Module 4 will assess the project performance to ensure the value from investors

Stage 4 Sustaining Value: Ongoing improvements and operations

- **Benchmark / bench-learning** – as cities move further into operations of smart lamppost systems there will clearly be a growing desire and opportunity for shared learning to: build market confidence for those cities that have not acted, and more generally for industry; drive further innovation, seek opportunities for further improvements, and the like.
- **Decision Support Tool (Module 4)**: Module 4 will assess the project performance to ensure the value from investors
- **Packaging Engaging Materials** – again a range of other supporting guidance materials for key city officers e.g. Market Case Studies
- **Case studies Template:** a template to collect the main insights from the experience to inform other cities

2 City Context

A city context is the central piece of the packaging triangle which links (i) societal needs (II) Technical Specifications and (III) business models and Financing together. Each city has different rules, typologies and demands which requires to design a specific solution tailored to the local needs. The three Sharing Cities demonstrators are significantly different. This section will seek to describe the framework to consider for the context of any city and explore the differences and similarities each city in Sharing Cities.



2.1 City Context Framework

The three Sharing Cities demonstrators are significantly different. Each city has different goals, geography and demands which sets the context to design a specific solution tailored to the city needs. There are some key local context elements that might influence the design and the deployment of the Smart Lamppost.

Key context factors to consider:

- **Political & Legal conditions**: play a strong to decide which solutions a city should select. Th city's experience on data and smart city projects has also an influential role. Political leadership and governance are also key to accelerate decisions.

- **Social & Cultural:** To determine which use cases could be upgraded to a lamppost the social environment is important such as the rate of criminality, the lifestyles and usage in electric mobility

- **Environment:** Climate, asset age and infrastructure, district type are all important factors for the smart lamppost allocation and landscaping.

- **Economy**: Budget of cities, economic growth, local market readiness for these smart lamppost technologies are also important considerations.

The Smart Lamppost is not a new concept and well proven and already deployed in several cities such as London, Paris. And yet the number of old assets in Europe is shockingly high.

2.2 City Context examples

CITY CONTEXT	Milan 	Lisbon 	London 
Political & Legal conditions:	• **Political leadership:** In Italy, Milan, the political leadership is vital to support the deployment of Smart Lamppost, the mayor need to be engaged formally to move forward opportunities. • **Governance** • **Smart city & data strategy** • **Citizens Engagement** • **Legal and Regulatory**	• **Political leadership** • **Governance** • **Smart city & data strategy** • **Citizens Engagement** • **Legal and Regulatory:** Certifications appears as a blocker for the city to connect external equipment to old lampposts.	• **Political leadership** • **Governance:** The governance of London is spitted between 33 boroughs which is unhelpful to deploy pan London smart lampposts as each boroughs owns their lampposts and therefore the decision to upgrade and in what forms • **Smart city & data strategy** • **Citizens Engagement** • **Legal and Regulatory**
Social & Cultural:	• **Security** • **Lifestyles** • **Demography** • **Cultural Heritage**	• **Security** • **Lifestyles** • **Demography** • **Cultural Heritage**	• **Security:** Some boroughs are high in criminality rates such as Camden, full light levels at night help to mitigate this challenge • **Lifestyles** • **Demography** • **Cultural Heritage**
Environment	• **Climate** • **Assets and Infrastructure** • **Natural Assets** • **Geography**	• **Climate**: The Lisbon climate is warmer and sunnier than the two cities which raises the question to consider other means to power a lamppost with renewables sources such as solar or Wind turbines. • **Assets and Infrastructure** • **Natural Assets** • **Geography**	• **Climate** • **Assets and Infrastructure** • **Natural Assets** • **Geography**
Economy	• **Market Readiness** • **Technology** • **Performance** • **Investment** • **Appetite for risk and innovation** • **Entrepreneurs ecosystem**	• **Market Readiness** • **Technology** • **Performance** • **Investment** • **Appetite for risk and innovation** • **Entrepreneurs ecosystem**	• **Market Readiness** • **Technology** • **Performance** • **Investment**: London as financial hub has an advantage when it comes to come investments, investors are eager to risk because of business tax rate deductions • **Appetite for risk and innovation** • **Entrepreneurs ecosystem**

3 Societal Needs

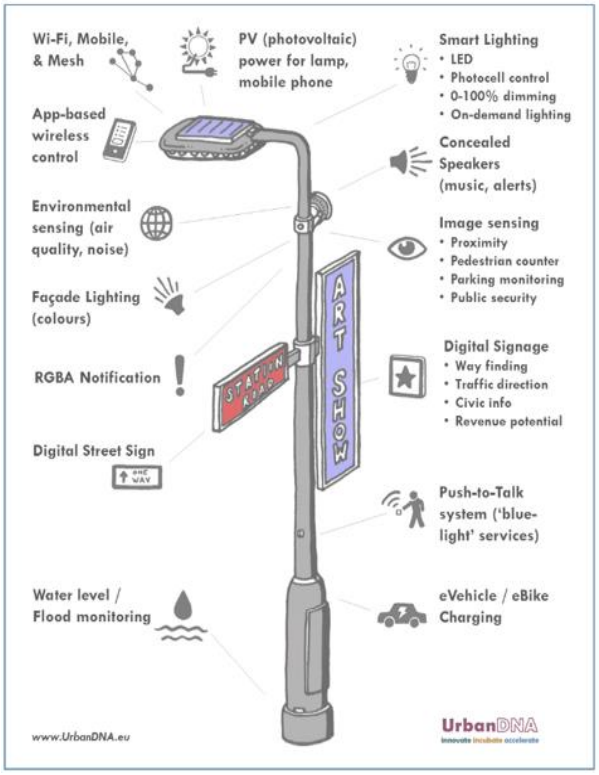
A solution has to address a need. That's the idea of societal needs to look at "why" do we need such solution in cities. This part will cover the different use cases that a Smart Lamppost can have and outline the different value.

3.1 Use cases

This section summarises the more typical light and non-light use cases that offer potential, in order to inform and then support selection of those that will be of interest. There are more than a dozen non-light use cases that have been identified, and these have categorised in six groups below. A Use Case template is included in Annex XXX that can be used to detail the specifics for those that a city seeks to move forward to analyse.

Lighting Use Cases

Application: A light-emitting diode (LED): device emitting visible light using less energy than traditional lighting. Lighting Control Management System (CMS) system designed to regulate the energy consumption to optimize energy usage, diming light intensity
Direct Benefits: Reduction in light pollution, energy and maintenance costs and carbon emissions
Wider Benefits: District attractiveness, Health, Safety
Barriers: Upfront investment cost



Energy Use Cases

Application: Electric Vehicle (EV) charging plug, Energy battery storage, Solar Panel Cell
Direct Benefits: Reduction in energy costs, carbon emissions, increase of electric car usage
Wider Benefits: District attractiveness, health
Barriers: Lack of proven business case

Public Safety Use Cases

Application: Speaker device for public announcements; Push-to-Talk button safety talk button on lamppost to contact the emergency line; CCTV, Public Alerts Posts and Info
Direct Benefits: Safety
Wider Benefits: District attractiveness, Safety, Health
Barriers: Surveillance issues

Environment Monitoring Use Cases

Application: Air quality sensors monitoring air pollution level; Water Level sensing to manage flooding; Noise sensor devices; Weather sensors
Direct Benefits: Data access, better info, better city planning and prevention

Wider Benefits: District attractiveness, Safety
Barriers: Weather variables

Signage Use Cases

Application: Public information and advertising on digital screens
Direct Benefits: Citizens engagement, revenues from advertising, public awareness
Wider Benefits: District attractiveness, engagement, business opportunities
Barriers: Lack of proven business case

Movement Monitoring Use Cases

Application: Parking and traffic movement monitoring
Direct benefits: Data access, better info, better city planning and prevention
Wider Benefits: District attractiveness, Health
Barriers: Right Allocation, Data Privacy Issues

Connectivity Use Cases

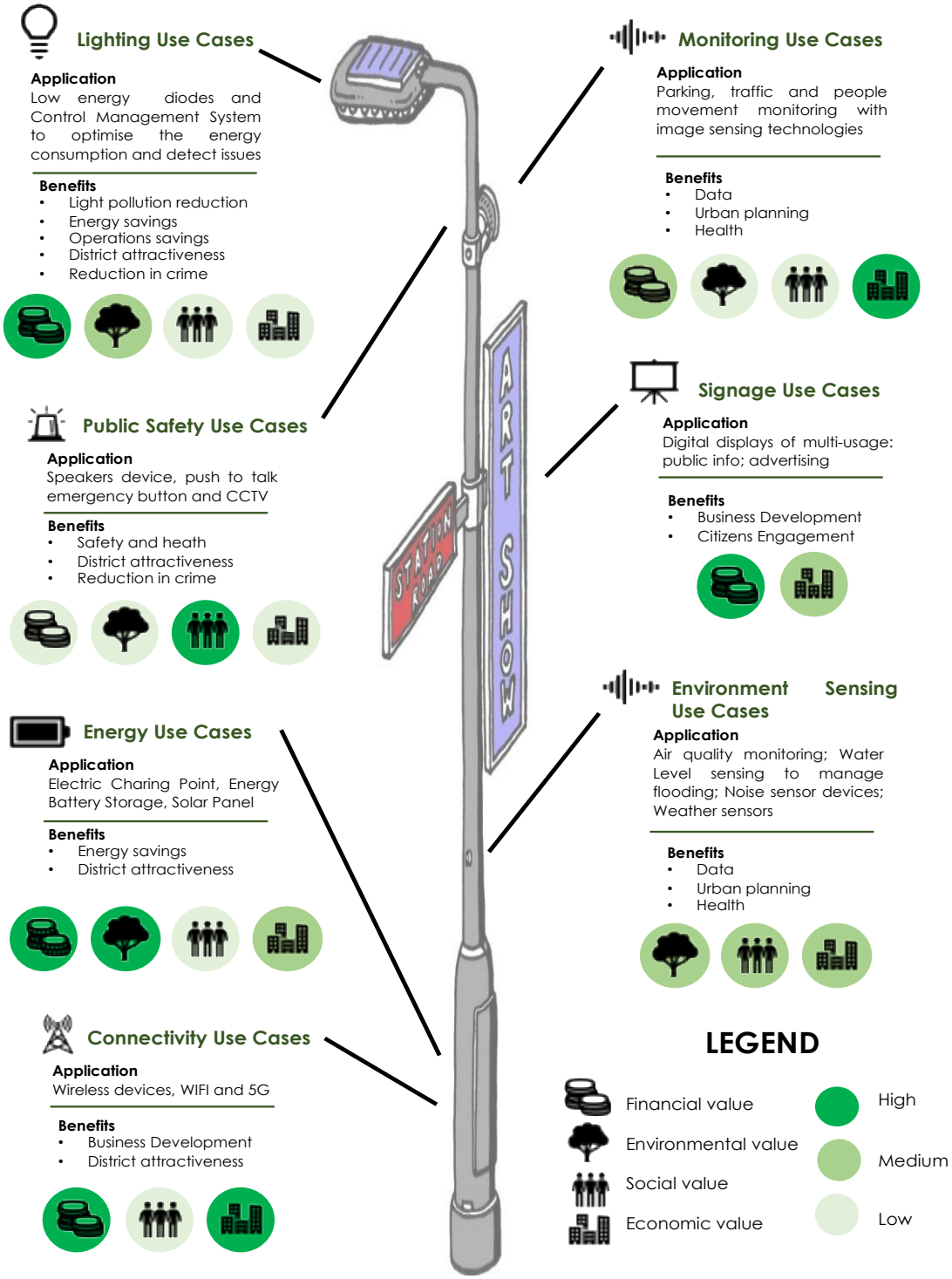
Application: 5G and Wifi and wireless communication with other urban technologies
Direct Benefits: Internet access

Wider Benefits: District

3.2 Value

The multi-purposing of smart lampposts can address pressing city needs, and offer new forms of value for communities and individuals. Of the multiple technologies and use cases to consider, each offer different forms and levels of benefit. The figure below indicates the types of value the various use cases offer considering four pillars:

- financial value (revenues, financial savings),
- environmental value,
- social value (health, safety) and
- economic value (local and business development).


Lighting Use Cases
Application
Low energy diodes and Control Management System to optimise the energy consumption and detect issues
Benefits
• Light pollution reduction
• Energy savings
• Operations savings
• District attractiveness
• Reduction in crime
Public Safety Use Cases
Application
Speakers device, push to talk emergency button and CCTV
Benefits
• Safety and heath
• District attractiveness
• Reduction in crime
Energy Use Cases
Application
Electric Charing Point, Energy Battery Storage, Solar Panel
Benefits
• Energy savings
• District attractiveness
Connectivity Use Cases
Application
Wireless devices, WIFI and 5G
Benefits
• Business Development
• District attractiveness
Monitoring Use Cases
Application
Parking, traffic and people movement monitoring with image sensing technologies
Benefits
• Data
• Urban planning
• Health
Signage Use Cases
Application
Digital displays of multi-usage: public info; advertising
Benefits
• Business Development
• Citizens Engagement
Environment Sensing Use Cases
Application
Air quality monitoring; Water Level sensing to manage flooding; Noise sensor devices; Weather sensors
Benefits
• Data
• Urban planning
• Health
ART SHOW
STATION ROAD
LEGEND
Financial value
Environmental value
Social value
Economic value
High
Medium
Low

4 Supporting Cities through the Project Value Chain

This section seeks to orient cities around the emerging toolkit and broader market factors that must be considered, and discuss readiness of cities, industry and investors to act.

4.1 Action at the Right Time in the Project Value Chain

Cities will typically go through a number of very conventional steps in upgrading / implementing their smart lamppost solutions, as shown in figure 3.

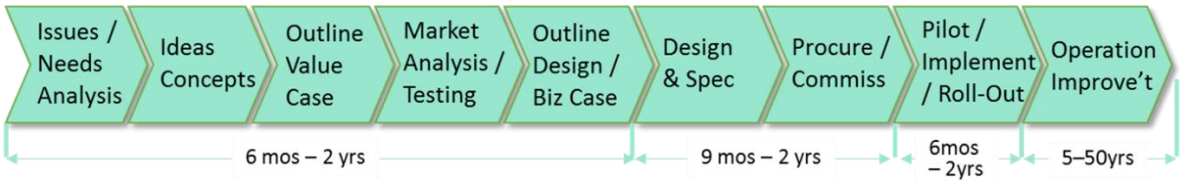


Figure 3 Typical Project Value Chain

Our opportunity is to help cities understand and address the opportunities that exist throughout this project value chain. The important point is that interventions should be planned in advanced and actions taken at the right time. Too often cities learn of what they could do when it is past the time to act, so lose out on opportunities for value. These opportunities may concern internal choices and needs (like awareness, financing, use case selection), or external (like aggregation of demand with other cities). Some of these opportunities are identified in figure 4.

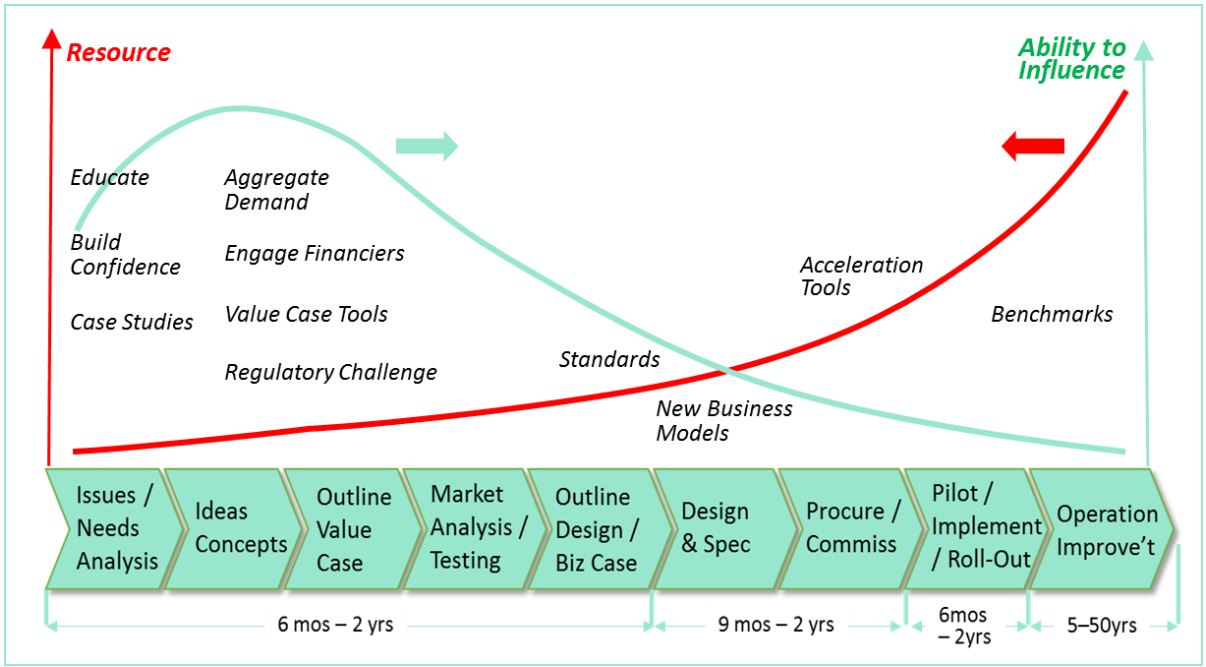


Figure 4 Potential to Influence through Project Value Chain

This document seeks to support cities to identify opportunities to maximize value ***early*** in the process, so that they can be acted upon in good time.

4.2 Tools to Support the Implementation Process

Whether a city is installing new smart lampposts, or (more likely) upgrading existing lighting stock to include smart services they will likely follow the natural logic shown in

the above figure. We have mapped the tools from the toolkit to the project value chain as an indication of what tools are presently available to help delivery (figure 5).

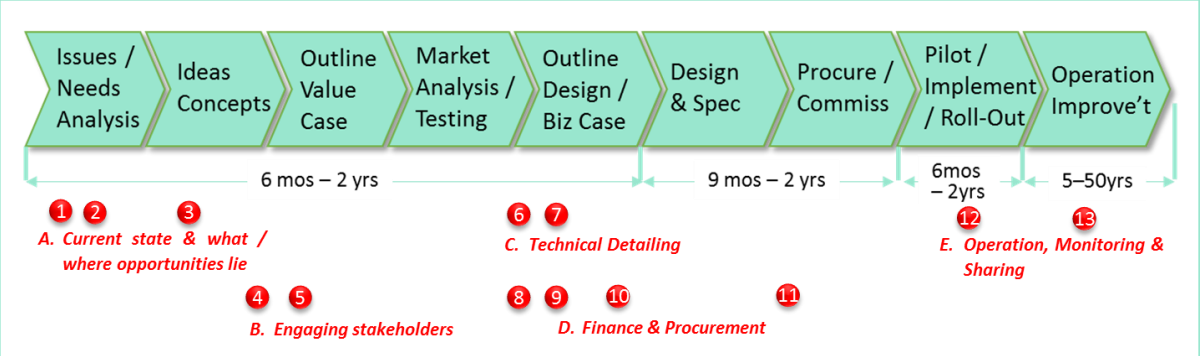


Figure 5 Toolkit mapped to Project Value Chain

The table below provides a step-by-step overview of each of the project steps identifying the tools that can provide assistance. There are naturally some overlaps and iterations that will take place in these activities.

Table 1 Mapping of Tools to Project Value Chain Steps

Value Chain Step	Description	Applicable Tools
Issues / Needs Analysis	Identify city lighting challenges and issues (likely to be community specific). These may be provider related (costs, assets, staffing, etc), or societal (safety, outages, light pollution). Capture the current inventory of lighting assets in a structured database (preferably gis), and ownership / operational accountabilities.	1.Maturity / Development Model 2. HL Asset Landscaping
Ideas / Concepts	Explore what the priorities are from a city operations, political and societal standpoint; recognising that some of the new potential from 'smart' services will be less known. Get more practical as to what specific use cases are desired in different communities, and compare the asset landscape with the future desires to determine gaps and challenges.	3. Use Case Templates 4. Stakeholder Mgmt 5. Ldshp Guide / Mgmt F'work (BSI)
Outline Value Case	Identify options / scenarios for upgrade. Estimate the notional financial and non-financial value from LED and 'smart' features	9. Decision Support Tool (coarse)
Market Analysis / Testing	Assess the local and international market for providers of solutions. Access learning networks to reduce effort and time required.	
Outline Design	Further detail the overall design	7. Detailed

/ Business Case	concept in different city areas. Identify risks; prioritise; seek to build in flexibility. Assess financial RoI, and non-financial case.	Specification 8. Tech 'Bill of Materials' 9. Business Case 10. Business Model & Financing
Design & Specification	Building on the asset landscape, agree the approach to specification (input or outcome based), and capture the design requirements. Consider opportunities for demand aggregation with other cities. Estimate costs and develop budget / financing plans. Evaluate business model options and select preferred model.	
Procurement / Commissioning	As appropriate to above, construct a suitable approach and documents to engage the market.	11. Procurement Templates
Pilot / Implement / Roll-out	Much of the technology is proven so question the need for and objectives of local piloting. Scale adoption can accelerate payback, and offer revenue potential.	12. Operational KPIs
Operation / Improvement	Identify a set of KPIs that support comparison over time for lighting / energy consumption optimisation; and more qualitative / strategic gains. Identify comparable communities / cities to share experience in a structured way.	13. Benchmarking / Bench-learning

4.3 City Readiness

The concept of city readiness is important. It is the basis by which scale adoption and demand aggregation can occur.

At a city-specific level, a city can make a coarse assessment of readiness through a status review of the value chain steps in the table above. This should consider technical readiness, and as importantly it must consider matters like: commitment of decision makers; societal preferences and engagement, availability of investment; capacity to act, and the like.

At an aggregated level, it is anticipated that the cross-SCC01 collaboration agreement, and EIP-SCC Marketplace can assist in the process of readying the market – demand side first, then supply and investors. (The task of societal readiness must rest with the city).

4.4 The Industry Perspective

The readiness of cities is paramount to market action. Whilst this is key, it is important also to consider the perspective of industry. And how that aligns (or not) with readiness of a city or cities (figure 6).

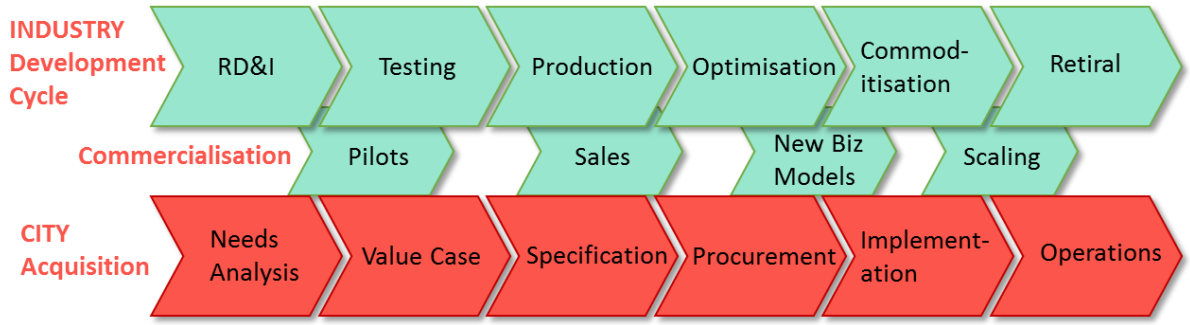


Figure 6 Alignment of City and Industry value chains

The smart streetlamp involves multiple components and several (new) industry actors. It also requires industry actors to work together in ways that perhaps they have not done before. This can introduce multiple challenges: relationships, capabilities, technical, commercial, size and influence, profitability, propensity to invest, R&D activities, etc. These factors should be brought into consideration. As a result, there will be many different motives and (dis-)incentives at work.

Additionally, the different elements of the smart lamppost are at different life-cycle stages in their development (technical readiness level: TRL). This introduces uncertainties, risks, and opportunities. A brief summary considering the major market segments is captured below:

- ***Lighting*** – new (LED) luminaires are now more mature and stable; costs have reduced considerably over recent years; further developments in lighting will occur however the basic benefits can be captured now. Enhancements in terms of integration of the luminaire with sensors and the like is still anticipated. Several large international players exist in the market, with new cheaper offers appearing from the east. A modest number of EU SMEs remain.
- ***Pole Manufacturers*** – the market is typically very fragmented with thin profit margins and thus limited capacity to invest. Market consolidation may occur. New designs to cater for the addition of equipment (load/wind load) and consideration of external and internal fittings is necessary and already occurring. Some very novel integrated component poles are emerging. Costs are also very variable depending on the scope of smart features and the extent wo which they are an integral part of the design. Challenges of consistency in design requirements, standards, and cost variance are anticipated. Additionally, for cities that are upgrading their stock without replacing existing poles, there will continue to be the need for elements of bespoke design and retrofit.
- ***Electro-technical Systems*** – this sector is fairly stable and though developments continue, there are no major changes anticipated that should hinder investment decisions in smart lighting upgrades.
- ***Control / CMS*** (central management systems) – field controllers, and central management systems are presently undergoing change. Notably, there is a trend to consider the extent to which CMS solutions should be or become an integral part of smart city (data) platforms. Cities should consult and be informed in this respect, and explore beyond the traditional lighting industry perspective.
- ***'Smart' equipment*** – sensors, transmitters, and other such equipment continue

to evolve significantly; as do their ability to perform more functions with smaller size and at lower cost. This can hinder decision making. Cities that hold back in taking action (even should it be proof of concepts) are likely to lose out on the resulting learning.

- ***Connectivity*** – considerable developments are occurring in terms of connectivity, and there is much discussion as regards the nature of connectivity needs of potential users of city services (notably eg automated vehicles). 5G, WiFi, LORA, LIFI are all connectivity means for multiple city services delivered from public and private sectors that are in flux.
- ***Data Brokers*** – data is without doubt becoming a far more valuable asset, and an array of smart lampposts provides considerable potential to collect and trade data. Data brokers tend not to be new players that deal only with the data; they are more likely to be existing providers that seek to play a greater role vis a vis data. The nature of data that will be collected, the valuation of it, and its monetisation, are all highly dynamic topics that warrant care and attention.
- ***Operations & Maintenance*** – cities are increasingly subcontracting this to the market. With the increase in smart features on lampposts there may be a capability shortfall in current operations and maintenance teams. Maintenance schedules and costs are also less certain, so this uncertainty should be factored into business cases.
- ***Power & Utilities*** – cities should evaluate this area carefully. Asset ownership, costs, energy consumption measurement, policies, and more all require consideration. Industry players are taking different interests and positions as regards smart lighting systems, and several of the larger players who see potential are bringing new business models to the table. These may come with conditions that, without careful due diligence, leave cities in compromised positions.

4.5 The Investors Perspective

Not unimportantly, the financial community are playing an increasingly important role in cities, as public budgets face growing strain. Typically, investors view cities as "too small, too slow, and too risky". Though a very broad brush statement there is some truth behind it. Investors are also often involved too late in the cycle such that they can neither contribute to the discussions on business models and financing where they could add considerable value; nor have the opportunity to be suitably briefed in terms of technical developments, challenges and risks. The result of which is that cities lose out, and the considerable funds that exist in the market are not targeted sufficiently towards public value investments.

There are of course multiple different types of investor (public grant, sovereign, impact, institutional, commercial, industry, VC etc), and all these have a different perspective and take different positions in the market.

There is therefore a real challenge to change the perspectives and views of investors towards (smart) cities in general and the smart lamppost opportunity specifically.

The humble lamppost can deliver a strong business case whereby by combining the certainty of solid energy and operational savings, with the modest (and riskier)

investment in smart services – some of which hold considerable upside revenue potential – an attractive investment case can be made.

Change in all sectors is required for the smart city and humble lampposts market to function more productively.

Cities should focus on:

- The opportunity to generate scale local, through individual scale action or demand aggregation amongst
- A "city-needs-led approach" with more common designs (at least logical, if not physical)

Industry should focus on:

- Collaboration between sectors and industry players to develop supply ecosystems
- Investing time to understand city needs better

Investors should focus on:

- Business models & financing structures that can incentivise action, and demand aggregation
- Getting involved earlier and developing greater technical understanding in order to contribute their skills more.

5 Inside the (Emerging) 'Toolkit'

This section provides further information on all the toolkit components listed in sections 1.5 & 4.2. The reader must recognize that these are at different stages of development and factor this into their use.

Cities are invited to apply the various tools described below within the logic and project value chain discussed in previous sections. Importantly, feedback on the use of these tools is requested to support the improvement mechanisms. The feedback process presently should be direct to Sharing Cities, and in time is anticipated to move more towards the EIP-SCC marketplace, and / or standards organisation processes.

Examples of their application, and experience from the Sharing Cities work to date is interspersed within this section, and included (& referred to) in annex.

5.1 Maturity / Development Model

Every city context and setting differs in many different ways: politically, regulatory, economic, social, physical, etc. However, cities can benefit from structured learning and comparison which requires a framework to support the process. And indeed, there will always be opportunities for cities to seek areas of commonality where joint activities can take place – be that design, market evaluation, or acquisition.

To assist a city in charting its improvement journey it will benefit from an honest appraisal of where it presently stands against a number of factors; and also set some ambitions.

Two frameworks are offered to assist this process.

The first (figure 7) shows three perspectives (technical, business model and finance; implementation) with 4 stages of ambition ('On-Ramps'). At each intersection, the model indicates a logical 'state'. This very coarse model helps stimulate debate and can help align functions within a city (and between collaborating municipalities). (Could update to align with below 5-stage model – indeed, with avail time)

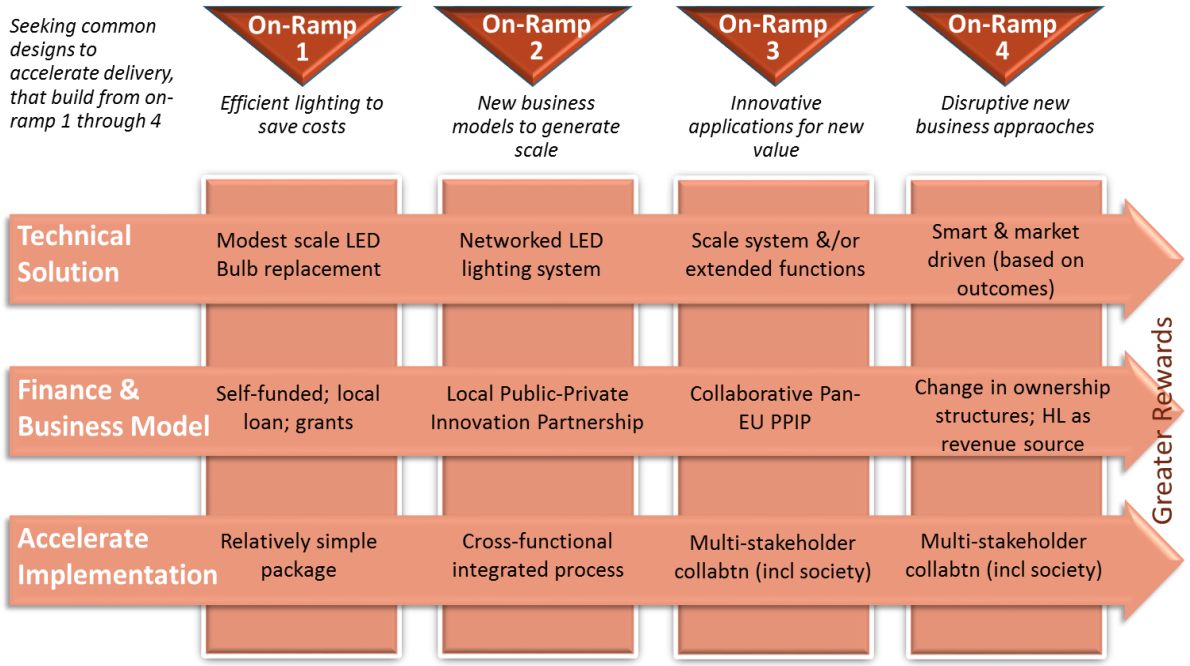


Figure 7 EIP-SCC Humble Lamppost On-Ramps

The second (figure 8) operates at a somewhat more granular level, and has been applied by Sharing Cities to assess the current state and future ambition (as a result of the programme). It operates by assessing a city against 9 factors and 5 levels. This is done by judgement, based on the indicative text for the present and desired state. In principle, the "Rules of Application" are that a city should seek to move 'one step at a time, and ensure they are not more than one-stage apart' during execution, to minimize risk of failure.

Humble Lamppost: 5-Step City Needs Assessment		Maturity Assessment Rating				
Element	**Goal** (what is a notional ideal leading practice future state?)	**On-Ramp 1: "Traditional"**	**On-Ramp 2: "Basic LED Upgrade"**	**On-Ramp 3: Basic Smart w/New BizM**	**On-Ramp 4: Adv'd Smart/New BizM**	**On-Ramp 5 Leading Practice**
Lamppost / Luminaire Asset	**Common logical architecture design at EU (EIP) level, supported by EU / International Standards Orgtns. Flexible 'component-based' solution that cities can select as appropriate (inter-operability). 'Future proof' design that industry apply innovation to. Affordable, and template based / easy-to-procure packaged proposition.**	Aging physical pole no / limited ability to take additional attachments. High consumption luminaires. High down-time; hi maintenance.				Component based smart lamppost, future-proof design for smart upgrades. Dynamic interoperability with city systems.
Societal view	Attractive lamppost design with pleasantly lit communities, with flexibility in operation (e.g. dimming artistic lighting etc). Safe place (e.g. 'push to talk'; brighten when danger). Optional Smart Services enabled (e.g. eVcharge; WiFi; Air..). Respect of personal data privacy (e.g. geofencing for retail offers).	Ugly old poles. Poor quality lighting. Sense of danger in some areas. Multiple call centre complaints		LED luminaire selected to suit location.		Community (residents AND businesses) budgeting and lighting / smart services SLA reporting.
Built Env't interface	Integrated bldng/place based lighting, with movement controls, safety and light level sensitivity. Auto-fault reporting. Smart services commissioned by local community groups. Façade and artistic / mood lighting where appropriate. Auto integration between 'place' and lampposts (eg power storage).	Badly placed street lights irritating residents (hi / low intensity areas). No integrtn between place & l'posts. Poles detract.		Light levels measure background building / retail light levels to set lamppost light.	LED luminaire selected to suit location (RGBA-W/Tri etc) mood /Façade lighting	Integrated building-place / lamppost lighting in design and operation. Mood / safety lighting.
Energy Mgmt System Interface	"*Energy Efficiency*": PVto power L/post. RE-FIT model to fund L/post. Monitoring. "*Store & Control*": Monitoring power consumption (accurate). CO2 emmission monitoring. Tarrif incentive. Storage (battery). Demand response asset. eV power; & eVto light. Licence life business model, revenue stream. Mobile phone charge. Store/Supply/Generate/DSR. Off-grid l'post. Micro-grid.	Street level energy measurement (often less).	No CMS. Measureemnt of energy consumption at area based level.	CMS (central mgmt system) for lighting level control by street / area level		Auto integration and light / energy optimisation PV/Alt energy integration systems.
Mobility interface	Mobility services; on/off & dimming as function of traffic flow. eSignage option. eV charge; eBike interface. WiFi mesh to better access eV charge network. Link with connected/assisted driving (infra-car etc).	Lampposts and lighting levels poorly integrat3ed with transpot system, with resulting risks.			eBike/Vehicle charge points with locatn-driver apps. Lamp to ITS interface.	Incentive / demand response posts to conditions, behaviours and other flows thru time-of-day
Data Platform	Defined sensor interfaces (gateways). Clearly selected (open) protocols. Defined interoperability layer (sharing). Identifies lamppost and lighting status. Identified use cases for data.	No systems support. Lamps set by mechanical timers, no information.		CMS integration with city platform.	Lighting / Lamppost-enabled data strategy (geofencing) to respect privacy & enable services.	Due proection of personal and asset data, yet innovative data services, with city / personal choice
Business Models Financing & Funding	Demand aggregation locally; networked internationally to enable best price / innovation mix; and support local SMEs. Business Models that incentivise aggregation; support x-silo delivery and liberate value. Multiple proof points for financial social ecological value.	Silo annual budgets that inhibit investment / long-term decisions. Constricting procurement practices.	Phase-based LED upgrade (little data aggregation)			Concession models. City-region commissioning. City-data monetisation for city/community.
Evaluation	KPIs: e-Charge (# charges per day; KW/day) Nos of useful 'push-to-talk' uses. KWhpv/KWh used. Nos Failures. Nos queries to answer	Outage complaints inefficiently handled. No/basic energy KPIs. No effective targets. Silo'd Ops costs.	Financial measures to demonstrate energy cost savings.	Public light levels set on top of background light levels to reduce public energy consumption.		Triple bottom line metrics, linked to commercial-retail monetisation / residential bonds
OTHER	Public Regulatory & Policy conditions that support swift commissioning / procurement; local business enablement and innovation.	Regulatory barriers. Constraining local policies. 'Silo' thinking and behaviours (almost encouraged).				Liberal regulatoiry environment that supports innovation. Open policy

Figure 8 City Humble Lamppost Readiness – draft *(full size version included in annex)*

In applying these coarse assessments cities should ensure honesty in assessment. They should recognise that these are draft and coarse assessments, however they are still useful. They should involve multiple people and perspectives in the process as this helps engagement and alignment – which will be important as the project advances.

This exercise can also be undertaken alongside the evaluation of Use Cases (ref section 1.3).

5.2 HL Asset Landscaping

Following the coarse assessment (above) the various stakeholders in the city will have started the alignment process, and begin to be exposed to the opportunities of lighting upgrade and resultant savings; as well as smart services on offer. A logical next stage is to understand in more detail what is in place within (sections of) the city in order to evaluate options and scenarios for moving forward. This we have termed "Asset Landscaping", and it provides a base-line of the present lamppost / lighting configuration on which to plan advancement.

An example of the Lisbon asset landscaping is included in annex 6.2.

5.2.1 Capturing the Current

Various databases will be available from several parties that capture the current state of a city's assets. This may be captured manually in multiple places, or electronically in one multi-purpose access-right-protected database. And as cities consider multi-purposing their lampposts it is likely that additional data will need to be collected than has been the case when considering lighting only.

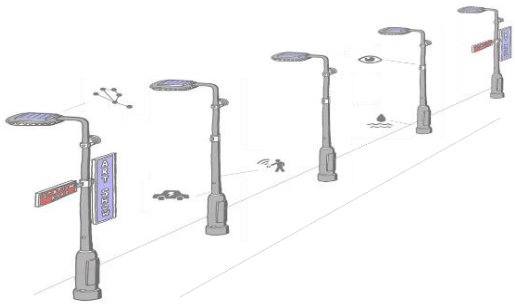
The first step is to consider what data to collect and what sources / parties to access this from. This process is clearly going to be very variable in time, however will likely take anywhere from 1 to 6-9 months depending on scale, state, and resources available to undertake the task.

Table 2 Asset Landscaping Data Collection (per Component)

Item	Data to Collect	Comments
Physical Lighting Assets		
Poles	Number, height, material of construction (wood, steel, concrete), coatings	
Lights / Luminaires	Number; % or # on old/new luminaires (& type: LED, SON, SOX, other); wattage; head type (% cobra, decorative,...etc.)	
Condition statement	Damage, vandalism, ware & tare – qualitative at asset specific level	
Ownership of assets		
Non-Lighting Assets		
Current smart service features	Sensors in place	
Systems & Operations		
Power system	Capacity; consumption; operation; outage;...	
Control system(s)	Field controller type(s); per light or grouping; manually set or remote;	

Power configuration	Cabling type; feeder box; controller set-up	
Other	Application(s) for capture/management of assets	
Financial		
Energy costs	Previous annual expenditure. Also by month; and where possible by area. Capture how this is monitored / calculated.	*Energy monitoring / billing can be so coarse it constrains smart upgrade*
Capital Plans	Previous annual capital expenditure (5-10 yrs). Book asset values.	
Operational Costs	Internal & external costs – annual; by type of work if available; by area; potentially by month/season	
General		
Location	Road type & name / surrounding environment / community	
Contracts	Contracts in place for operations / maintenance - type, length etc	

5.2.2 Defining the Future



Having base-lined the assets, the task is now to develop plausible options for upgrade plans. This will likely result in different levels of internal and external consultation, which should be considered and planned for at the outset.

It is likely that in a city there will be some existing policies and plans for lighting (and possibly smart services) which will be area based. These needs can be captured dependent to the specific areas: e.g.

- Central business district
- Industrial areas
- Special areas – like events / stadia etc
- Old historical
- New developments
- Inner residential (high rise – social housing vs non – private)
- Outer city residential
- Ribbon developments
- Other considerations, like safety, high risk areas, etc.

Where new-build or replacement columns are planned, consideration should be given to whether columns are indeed needed (e.g. locate lights on other fixtures).

Where luminaires will be upgraded and smart features installed consideration should be given to what density will be required for each of the smart features (i.e. how many air quality sensors will be required in any particular location, at what levels, measuring what particular features of air quality – particulates, chemicals etc; or what type of connectivity service is planned and what array of transmitters is required – also affected by vertical obstructions from buildings).

This process, for different types of smart service, is still typically in the learning phases within cities. Cities should seek to learn from others and plan for their own

experimentation (piloting where necessary; proof of concepts certainly; and eventual roll-out).

The table below provides a rough guide to density of the various sensors and other equipment that a city may wish to deploy.

Table 3 Initial Smart Equipment Density

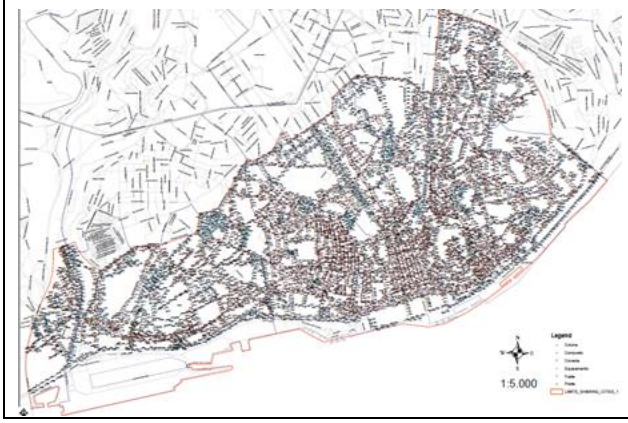
Sensor / Equip't	Density	Comments
Air quality	Consider 1 in 100 columns in air quality hot-spots	
Noise		
Water levels	Culvert monitors	
Connectivity		
Parking Video surveillance	Videos typically cover ~30m radius	
eV charging	Highly variable. Consider 1 in 100 columns in target areas	*Consider fast or slow charge, bike/car charge; costs, current & future needs. Consider other charge locations (e.g. in buildings/malls etc). Public / private facilities. Recommend gis mapping to support process.*
Energy storage		
Geofencing	Likely in central areas or special locations (eg stadia, retail)	
Push-to-Talk	District dependent	
Public signage	District dependent	
Speakers	District dependent	

5.2.3 Tools to Modernise and Sustain the Process

The process of data capture, particularly for large cities with large numbers of assets, involves significant time and cost. Consideration should be made to employ digital means to reduce the burden and improve the quality of data capture: handheld devices for field crews; applications that support asset capture; installation of QR codes on columns and equipment to also support ongoing data maintenance and the like.

An example of how the HL Asset Landscaping aspect of the toolkit has been applied to date in the Sharing Cities programme and lighthouse cities is given below, and a more detailed account of asset landscaping in Lisbon is included in annex 6.2.

Applying the toolkit in Sharing Cities: HL Asset Landscaping





5.3 Smart Use Case Templates

The table below shows the use cases captured by the Sharing Cities work and the DIN specification.

Table 4 Use Case Listing (ShCiteis & DIN)

Humble Lamppost Use Cases		DIN	ShC
LIGHTING Related			
L1	Basic LED Energy, GHG, and Maintenance Improvement		Y
L2	Additional Energy Savings / Optimisation (CMS "Trim & Dim")		Y
L3	Safety, Attractiveness & Façade / Mood Lighting (incl Improve the Quality of service and people safety – fault reporting)		Y
L4	Alternative Clean Energy – PV cells to power some lights		Y
NON-LIGHT Related			
NL1	Event-controlled adaptive street lighting system	Y	
NL2	Traffic monitoring (Driver Information; Traffic Monitoring; Parking)	Y	Y
NL3	Intelligent communication between vehicles and the imHLa	Y	
NL4	Public Wi-Fi	Y	Y
NL5	Wireless network support with a picocell or microcell	Y	
NL6	Environmental data acquisition (air quality, noise, water levels, etc)	Y	Y
NL7	Charging station for electric vehicles (vans, cars, bikes)	Y	Y
NL8	Drone charging infrastructure	Y	
NL9	Energy storage	Y	

NL10	Remote maintenance	Y	
NL11	Public security	Y	Y
NL12	Private security and surveillance / geofencing – pedestrian monitoring	Y	Y
NL13	Signage and advertising	Y	
	Public Engagement (safety of place, information speakers, information signage, tourist information)		Y

The DIN standard explicitly excludes lighting only use cases.

As these use cases have been developed independently there is a need for some future alignment. The DIN use cases can be found in specification *PAS 91437 "imHLa – integrated multi-functional Humble Lamppost"* (to be purchased from DIN). The Sharing Cities use case template and an example is included in annex 6.3 & 6.4. The latest set is available in D3.11.

Figure 9 provides a visualization of some of these smart services for the humble lamppost.

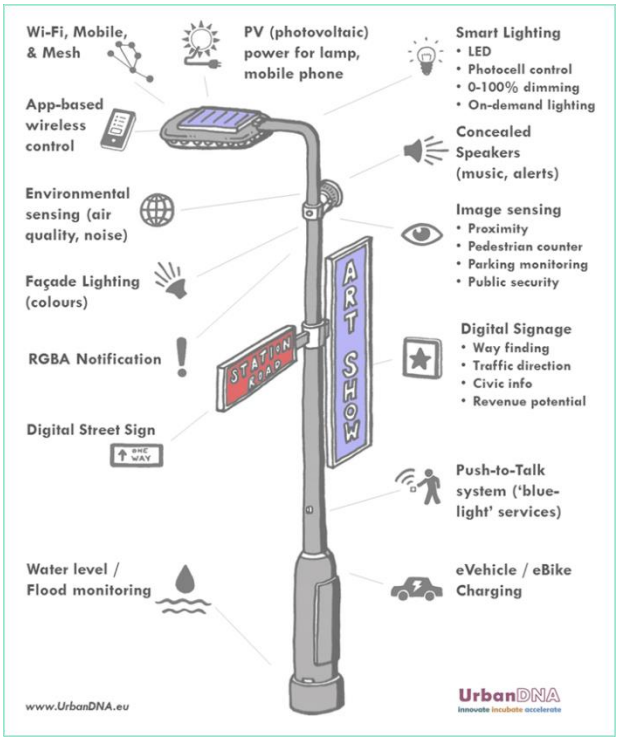


Figure 9 Humble Lamppost Potential Use Cases

Each use case will require a different set of equipment and services. Some of which however serve more than one use case. This relationship needs to be understood not only at an individual lamppost level, it requires consideration at a city-wide level – as the configuration and density of smart features varies (i.e. how many sensors are required per number of poles).

The DIN specification additionally considers the overall solution at different perspectives, specifically:

- Physical asset / architecture
- Functional
  - Electrical integration
  - Communications integration
- Connectivity
  - Digital integration
  - Data integration

Integrated multi-functional lamppost – integration of digital services This relationship between use cases (functional) and communities (physical), and the overall system (logical) is something that is still in the market learning phase and will continue to see ongoing 'trial and errors' and innovations. It is impacted by multiple factors – including importantly how society responds to these new features. As such there is and will be

no standard approach. There will however be plenty of scope for shared learning – particularly between cities.

To deliver the use cases identified above requires considerable cross-functional collaboration within cities, between cities, and between sectors / service providers. Mechanisms must be established to enable this, and allowance must be made in city project plans to cater for this. This is also particularly why the leadership guide and management framework documents are beneficial.

An example of how the Use Case Template aspect of the toolkit has been applied to date in the Sharing Cities programme and lighthouse cities is given below, and a more detailed account of use case selection in Lisbon is shown in annex 6.4.

Applying the toolkit in Sharing Cities: Use Case Template

Cities worked together with the task lead to consider uses for the humble lamppost, developing common use cases; these can be found in annex 6.4. These use cases were then developed and personalised to each city depending on interest or which were considered to be more applicable.

Initial UC focusses taken forward by each city:

Lisbon	*UCs in Lisbon focus on:* - *Light- LED conversion in 2017 ; LED & CMS systems in 2018* - *Non-light- Environmental monitoring (Air quality); traffic monitoring*
London	*In Greenwich, the UCs have been linked to RBG priorities, area context & resident needs. They focus on:* - *Light- CMS systems to couple with LED* - *Non-light- Environmental monitoring (Air quality); Smart Parking*
Milan	*In Milan considering area, residents other partners. Initial choice was for:* - *Environmental Monitoring Services* - *Public Engagement using Lampposts* - *Transport & Mobility Services* - *Indoor Monitoring Services*

5.3.1 "Bundling" of Smart Solutions

The Humble Lamppost is considered an obvious quick win for cities. In implementing the various HL use cases, cities may consider plans for associated smart services.

Figure 10 shows some of the other assets and services that cities may wish to align with smart lamppost implementation plans. Given the nature of these other services, this reinforces the need for cross-functional collaboration. It may also involve work to align business cases, and consider joint business and financing models.

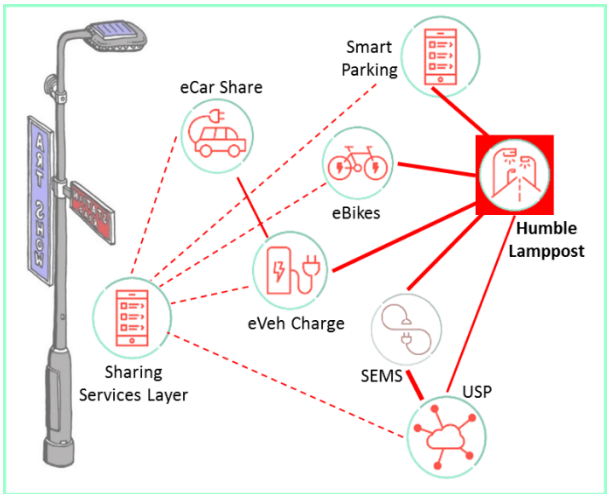


Figure 10 Complementary services to the Humble Lamppost

[NB SEMS = Smart Energy Mgmt Systems; USP = Urban (data) Sharing Platform].

5.4 Stakeholder Management

Traditionally the lighting engineer would be the lead person to decide what happened with city street lights. The scope of smart lampposts, however, spans much wider than lighting (depending on what use cases are selected) so a significantly larger number of service owners need to be involved. This, and the fact that smart lampposts are often an early step for cities into 'smart cities' tends to attract political attention; and may also warrant community consultation.

This broadening of parties involved warrants greater attention to stakeholder management, as there are several considerations that are raised that need debate, like:

- Who pays for the smart equipment – both capex, and maintenance?
- How should the costs and different forms of value, delivered over different timeframes, be managed between service lines?
- Are the capabilities in place to acquire and manage the new system? And where should these capabilities lie?
- Which smart services should be included, and in which parts of the city?
- Who holds the mandate for decisions on smart lampposts?

Potential stakeholders may include:

Within 'city hall'

- Lighting engineer – traditional lighting accountabilities
- Transport department – road safety, eV charging, signage
- Environment department – air quality, water monitoring
- Community Engagement / Communications
- Smart City lead
- Chief Technology Officer
- Policy Advisers
- Politicians

Others

- City communities
- Utility provider(s)
- Financiers
- Service operations and maintenance providers
- Lighting manufacturers
- Telecoms providers, Retailers, etc

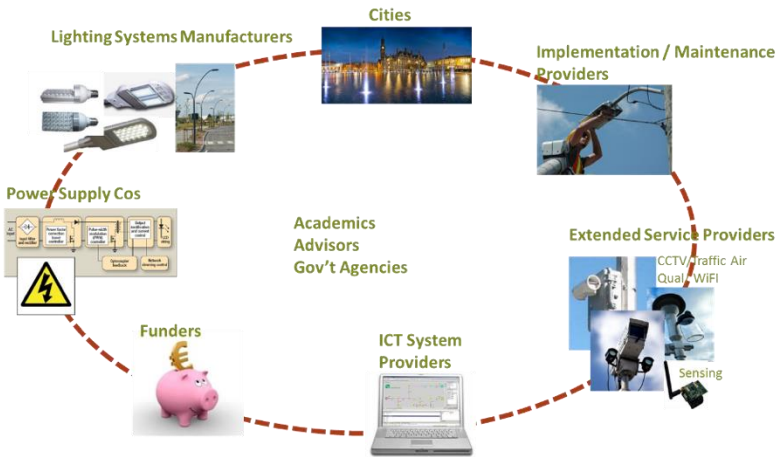


Figure 11 Smart Lamppost Actors

Consultation and workshops with these stakeholders will be required. Some may be unadvised, disinterested, or dis-incentivised to engage, so the value case will need to be explored from different perspectives to achieve a balanced and aligned decision.

Stakeholder mapping and management tools may be required. An example of which is shown in figure 12. Stakeholders can be broadly separated into three categories of

importance, and perception or received opinion of their position vis-à-vis the smart lamppost initiative can be formed. By understanding the underlying reasons for that position and taking appropriate measures, the various stakeholders can be aligned – which means bringing sufficient into the 'zone of critical influence'.

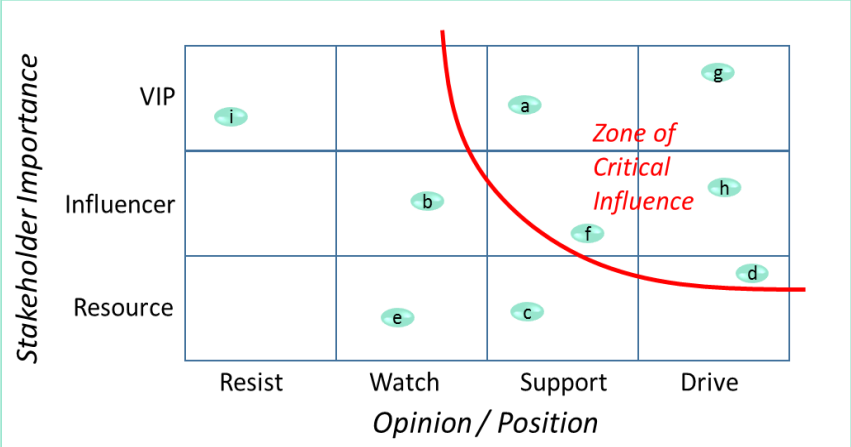


Figure 12 Stakeholder Management Tool

An example of how this part of the toolkit has been applied to date in the Sharing Cities programme and lighthouse cities is given below.

Applying the toolkit in Sharing Cities: Engaging the relevant stakeholders and checking against the cities priorities

All cities involved in the Sharing Cities project made a commitment to explore and pursue the introduction of the Humble Lamppost, and to realise its benefits, at various scales. However, implementing smart lighting solutions even if commitment exists in one part of an institution, still requires the briefing, integration, management and often convincing of additional stakeholders at an early stage. It was recognised that although this sits as a secondary part of the toolkit to 'considering status and where the opportunities are'; it is beneficial to engage stakeholders from the outset, particularly those stakeholders who may benefit or use the data coming from the smart lighting solutions.

5.5 (BSI) Leadership Guide & Management Framework

City political leaders are typically unadvised about the key opportunities, challenges, and leadership needs relating to smart city solutions. The smart lamppost is no exception.

Department and Service Heads in cities (and in some case with their service providers) struggle to work across service 'silos' to benefit from synergies; for multiple reasons.

BSI (and ISO) is in the process of delivering two sets of publications to help resolve the above and significantly assist stakeholder management particularly within city administration:

- Leadership Guides, that are short (6 pages) key issue documents to inform city leadership about what their steps could be to help promote action
- Management Frameworks, that provide an organising frame for multiple

functions to view a topic through to help promote understanding and alignment.

This document falls in the category of the second. Sharing Cities will draft a leadership guide to complement this. Both are anticipated to be submitted to BSI for adoption / adoption into national and in future international guidance.

5.6 Market Research and Analysis

Supply markets and pricing differs considerably across European cities, and it is important to understand these variances, as they offer opportunities for the city, and also for local SMEs – often a priority for cities to support.

Industry analysts can support this process, and there are several reputable analysts that have studies the smart lighting market. Typically, they will address only a small number of the larger EU markets in detail. This is not so important for the components that are predominantly supplied by major international companies.

However, for specific elements of the overall smart lampposts solution it will be important to evaluate local market developments. For instance, for lampposts, sensors, digital signage, or connectivity solutions – and in some countries for innovative lighting fixtures (which are typically supplied by major international players). Cities that have implemented solutions recently; and city associations / networks provide sensible means to access market intelligence and should be exploited to save time and resource. Market analysis can benefit from considering the various elements of the solution (see 'Bill of Materials').

An summary of Lisbon's market research is included in annex 6.5.

5.7 Detailed Specifications and Standards

Cities need to specify their requirements for smart streetlamps, however the familiarity with and availability of suitable standards to support this process is inadequate. As a result, cities often rely on engaging consultants / advisers to support the process; specify their needs on a best endeavours basis; or rely on industry to advise what is most suitable. This approach is deficient when one considers *smart* streetlights, as:

- Cities need to be more deterministic about what they seek as regards the 'smart' features, so must be fully aware of options;
- The market landscape is fast changing and few recognised industry norms exist (this therefore risks 'vendor lock-in');
- Engaging advisers on a city-specific level (particularly for small and medium cities) is a costly and time-consuming venture.

Considering the status quo of standards; there are multiple standards published on road and street lighting, issued by a large number of Standards Development Organisations.

Annex 6.5 provides a summary of the situation (as of 2015). This analysis provides an excerpt from the ISO Smart Cities Strategic Advisory Group (SAG) report, which included a "deep-dive" analysis on the smart lamppost.

The key messages from the SAG report, and the more specific deep-dive analysis are:

General SAG Report Points

1. *Cities are not usual users of standards. They often perceive standards as documents that constrain (regulation or compliance requirements), rather than provide guidance and best practice (which is also the case). They also often assume that their advisers, providers and suppliers will use the appropriate standards, rather than take an informed stance – often through resource constraints.*
2. *The recommendations to ISO from the SAG (which are presently in discussion / action) include to develop three tiers of guidance (i) Leadership Guides (ii) Management Frameworks, and (iii) Technical Standards. The former two are of particular relevance in a cities context. Item (i) seeks to inform political and senior city officers about opportunities and key issues to be aware of. Item (ii) seeks to help inform the various 'heads of services' and cross-cutting functions about the needs and advantages of working across agencies, departments and functions. Both are therefore biased towards the (less-frequent-user) city audience.*

Specific Smart Lighting Points

3. *A number of observations were made:*
   - *There are a large number of detailed technical standards available; and it would seem few documents that might be considered Management Frameworks [therefore a gap]*
   - *Most standards are national; none are ISO; and few are international.*
   - *There are very few standards that look to system performance (financial / GHG benefits etc)*
   - *No standards or guidance appear to address integrated infrastructures ('smart' services)*
4. *Recommendations made include:*
   - *Develop a few leadership guides that make the opportunities that are very apparent clear to city leaders*
   - *Develop a number of more business-focused 'management frameworks' that can help accelerate decision making and action in the market – addressing the identified gaps above (e.g. solution architecture; business case; business models).*
   - *Address systems interfaces – physical connectivity, communications, & data – through technical standards*
   - *In recognition that the existing materials are predominantly nationally produced; and particularly for the upper two tiers of document; consider development of these at an international SDO level; best perhaps as multi-logo documents.*
   - *Consider rationalising the existing portfolio of standards for this high value topic that offers rapid gains*

The second portion of Annex 6.5 provides an inventory of the EU country (& some international) standards. These have not been classified in terms of specific relevance. However cities can at least ascertain, based on the list and observations above, which of these documents are more likely to be relevant to their implementation plans.

Some industry associations are observed to be working on specifications and guidance documents, and these may be relevant to read. An example being the ongoing work of TALQ.

This publication is anticipated to provide the base material for the development of a 'Management Framework'; and a Leadership Guide is planned to complement this.

As noted earlier, DIN has recently published a standard PAS 91437 "imHLa – integrated multi-functional Humble Lamppost". This provides details on non-light use cases and

considers five different light configurations. This is being translated and will be published in English in mid-2017.

5.8 Technical "Bill of Materials"

The major components of the smart lamppost solution include:

1. Pole
2. Luminaire
3. 'Smart' Fittings
4. Electrical System
5. Control System
6. Civil Engineering

More details on the above are covered in section 2.

An example of how the above aspects of the toolkit which make up the Technical detailing section of the toolkit has been applied to date in the Sharing Cities programme and lighthouse cities is given below.

Applying the toolkit in Sharing Cities: Technical Detailing

All cities have begun the technical detailing section of the toolkit; where we start to establish what is necessary technically to bring the use cases and humble lampposts to life. Each city has conducted market research, and identified particular technical aspects which need attention or they need to overcome. Particular strands of technical detailing have included: the enclosure, sensor acquisition, and power requirements, as is described in the main body of this deliverable document. City specific examples of these are given in the table below. Enclosure refers to the casing around the devices that are often deployed.

Lisbon	- The majority of the lampposts in Lisbon's intervention area are street wall lampposts (consoles). Because these types of lampposts don't have a pole, it can be challenging to try and attach a sensor system without disrupting their aesthetics. - The electrical system in Lisbon, on lampposts that don't have CMS, is described as being on/off, which means that as soon as the power is on, the lights also go on. This fact implies that this lamp posts are only powered during the night, not having electricity the rest of the day. This brings some implications on services that need to be powered throughout the entire day (e.g. Wi-Fi). However, in lamp posts already converted to LED with CMS would be possible to supply 24h continuous voltage, overcoming these challenges. - One of the objectives is for devices to be sustainable, meaning that it would have means to keep operating without the need of being plugged continuously to the electrical grid. This needed technical consideration of a battery system. To have a better idea of what battery to use, in terms of its capacity, a sizing of the system's current energy expenses was conducted.
London	- Much time was spent conducting market research and initial market engagement into the technology (sensors and devices) that could be used to deliver the UCs, we found fast changing market with different technology being brought on line at pace. This made it difficult to determine the best devices deploy to implement the UC in an effective and useful way to the city. For example in considering air quality monitoring as part of the environmental sensing UC, the devices deployed need to add to the Borough's existing monitoring network in a reliable and accurate way. We found that there is a quality versus price issue, in that it is quite difficult to find a good quality sensor at a reasonable price. - Regarding the lighting UC of CMS/LED, it was important to consider how the technology could be managed, scaled up, and integrated with other hardware. This consideration has driven the trialling of multiple systems and in 2017 we will trial 2 types of CMS system, one is a mesh system and one is a system where one individual node can be added without the need to be accompanied by others.
Milan	-

5.9 Business Case

LED replacement offers a strong return on investment, with 50%+ savings on energy consumption and ~50% savings on operations and maintenance costs. Payback can be in the 4-8yr range. As a result, cities are actively upgrading their lighting stock to LED.

The opportunity is to use the compelling savings case to stimulate investment in smart services at the same time, thus reduce re-work on assets and design in smart features at the outset. This also reduces the barrier to business case for smart services only (particularly when they involve justifying non-financial benefits). At present the percentage change across Europe is still modest (10-20%), so the opportunity to do both LED upgrade and implementation of smart services in parallel is very real, and it is now.

The financial and non-financial value from smart services using the lamppost as a multi-purpose asset varies with ambition. Figure 13 illustrates this, by considering 5 stages of development from LED upgrade only, through optimisation of light by applying smart control systems ("trim and dim"), implementing safety (e.g. CCTV; push to talk) and environmental sensing (e.g. air quality), to 'smart' services (e.g. eV charging, connectivity, digital signage), to data trading through for instance geofencing pedestrians. The financial benefits for environmental and safety improvements are hard to demonstrate and tend to be considered as a cost. Conversely, the potential revenue upside (for the city or industry) from level 4 or 5 services can be substantial – and indeed greater than the financial case for energy. The upper levels are however less proven, potentially riskier, more innovative, and business models are less mature for these. Focusing on levels 2 and 4 would seem to be best from a financial business justification standpoint.

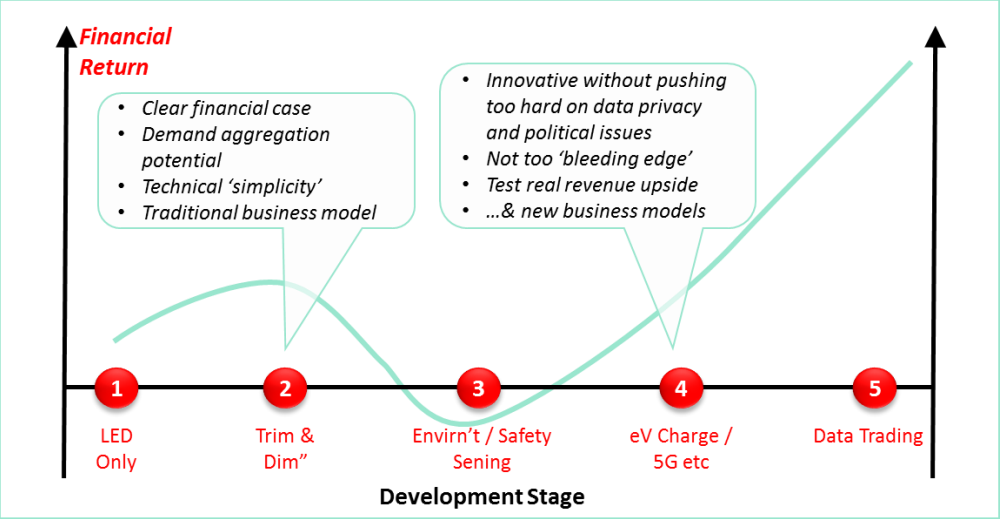


Figure 13 Variable financial returns from different levels of 'smart' ambition

5.9.1 The Challenges of Different Forms of Value

Those involved *only* in lighting and energy may not be incentivised to consider additional smart features, as they incur extra effort, complexity, and risk. Or benefits that fall in other service lines. And potentially with different (longer) timeframes (e.g. health benefits from better air quality).

Different stakeholders will be motivated to focus more on one side of the equation than the other, and perceive and value benefits in different ways. It is important to respect this, and provide the appropriate analysis to support these differing views. Multi-criteria analysis (with suitable weighting and ratings) may be used to help bring these different views together.

These variables include:

- *Costs*: capital, operations and maintenance (people, energy, materials)
- *Financial benefits*: capital / cost avoidance, operational savings, revenue
- *Other benefits*: speed of response, flexibility, safety, community satisfaction, visual appeal, reputation, health

5.9.2 The Case for Collaboration and Demand Aggregation

Collaboration to share experience and learn, particularly given the relatively new are of smart services, is clearly of value to explore.

Collaboration between districts within a city, or amongst cities can lead to considerable additional financial and other benefits, through demand aggregation and economies of scale. This can also help de-risk or share risk between cities that test and prove different smart features. Without doubt, the supply base will be more attracted by significant volumes coming to market. The perspectives of demand and supply side as regards volumes and value is indicated in figure 14.

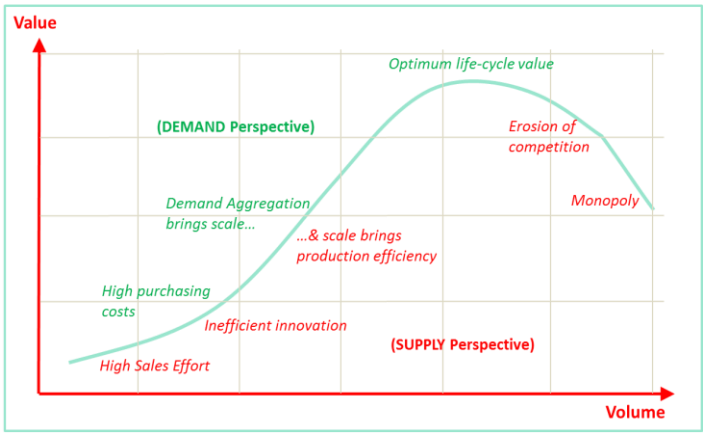


Figure 14 Economies of Scale - Demand and Supply Perspectives

The key question is what is the optimum volume to bring to the market for a smart lamppost implementation? There is no known empirical evidence available on this topic, however we have shown the contrast of a few potential acquisitions in figure 15, which would suggest that the best value for purchase of smart lampposts may well be at volumes that are significantly greater than most cities typically procure at – which is typically in the order of (low) 10s of thousands – with a few notable exceptions.

Clearly this is a coarse analysis and several factors must be brought into consideration.

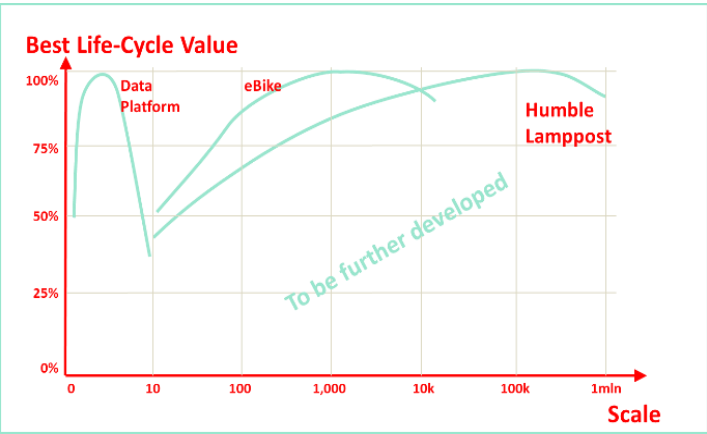


Figure 15 Seeking optimum economies of scale

Certainly, collaboration on design amongst cities (i.e. greater levels of commonality – standardisation) will lead to reduced costs; even with no joint procurement. Benefits will likely only accrue with joint procurement.

Given the different maturities of the smart lamppost component markets, and the desire to ensure an open market for (local) SMEs, cities may seek to stipulate specific conditions in procurement text concerning various sub-components of the solution.

5.9.3 Developing and Appraising Options

It swiftly becomes obvious that there is no simple singular route forward for a city, and that the development and evaluation of plausible options or scenarios is important. These may consider different scales of project (from a community within a city, to

collaboration between cities); and/or different use case combinations; and should consider the different risk profiles and levels of market maturity.

Within each option, it is recommended to perform some sensitivity analyses (e.g. on costs, benefits, expected product life, etc.). A consistent approach, with proven tools aids such work. Two tools are discussed below.

5.9.4 Smart Cities Council (SCC) Decision Support TOol

The Smart Cities Council (SCC) developed a decision support tool for smart streetlighting, which is freely available for use. It enables a city to capture the key data for its lighting assets, and provides a reasonably in-depth analysis of energy savings. What it also seeks to do is identify the costs and benefits of a number of the more attractive non-light use cases.

The tool (figure 16) was developed in collaboration with the EIP-SCC Humble Lamppost initiative. In developing the tool, it became apparent that costs for the non-lighting assets were both variable and not frequently revealed by industry. Additionally, factors in set up of cities as regards their lighting assets, and associated costs demonstrated the need to be able to set a number of parameters at local levels. Further collaborative development work is desired within the EIP-SCC to test, and strengthen the quality of the tool; aid learning; accelerate and improve the confidence of decision making.



Figure 16 SCC Smart Lighting Decision Support Tool

5.9.5 Scottish Futures Trust Tool

The Scottish Futures Trust (SFT) has developed a very detailed tool that addresses core LED lighting savings only. This is also available at no cost. It does however require considerably more data input to perform the analysis.

5.9.6 Other Tools

Other tools are available; some developed locally (and are thus not known or captured here); some developed by specific industry players. The quality and reliability of these should be carefully reviewed if used to ensure no bias.

5.10 Business Models & Financing

Smart city developments have, to date, relied extensively on grant funding, and industry investment or incentives. Smart has not, as yet, become 'mainstream' from a financial justification standpoint.

This must change.

A survey of 25 EU cities[1] revealed how they expect to achieve their smart city ambitions (figure 17).

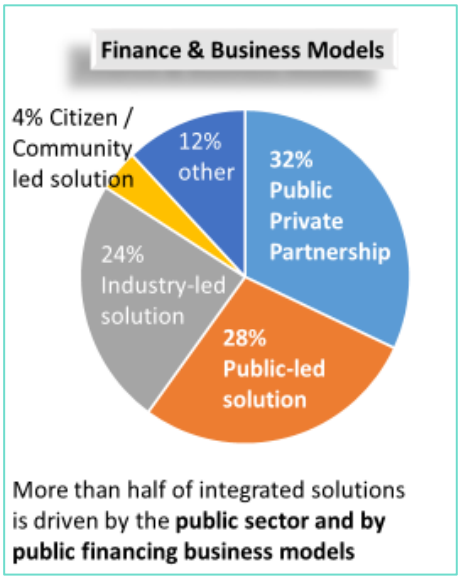


Figure 17 Financing by which cities anticipate achieving smart ambitions

Because smart lighting offers such a compelling return case, there are real opportunities for cities collectively to demonstrate the savings. The result of this would be far greater attraction of market funds to support scale adoption of these solutions. This would help counter the current reliance on public funding.

This trend has been observed of late in some cities, where service providers (e.g. that hold long term contracts), utilities, ESCOs[2], or investors have been seen to propose apparently attractive commercial terms to cities for taking ownership of lighting assets, or a long-term concession. The advantage of this is that it can lead to rapid scale upgrade of lighting systems, where constrained public budgets may well have precluded this.

Investors that offer attractive terms may well also see upside financial potential which goes beyond the returns available from energy savings. These can be in the form of revenue streams from WiFi, advertising, eV charge, etc. Any contract that limits flexibility for cities or providers to influence the future delivery of public good services may be an unhelpful constraint, so in seeking financing (and thus agreeing business models), cities should carefully evaluate the relative merits of these options, and ensure flexibility is built in.

5.10.1 Business Models

Business models address conditions of ownership, investment, return, accountability for operations and performance for provision of a service to a prescribed customer base. It is useful to consider the value chain elements of design, build, operate, maintain, and asset ownership. Thus, terms and models like DBO (design, build, operate); BOT (build, operate, transfer), BOOT (build, own, operate, transfer) emerge

[1] *From "Analysing the potential for wide scale roll-out of integrated SCC solutions", report to DG ENER*
[2] *ESCo = Energy Savings Companies – a commercial or not-for-profit business proving a range of energy savings solutions, and often with some form of public involvement*

in terms of the accountability and risks that various parties / consortia take through the life of a project or service. DBO, BOT, BOOT tend to be used in the context of multi-party often public-private-partnership (PPP) models. New models where society get involved are emerging, aided by digital communications and capture means: crowdfunding or PPPP (people-public-private partnerships).

The introduction of new 'smart' services and new potential revenue streams from multi-purposing lampposts offers scope to consider several new ways of treating ownership, investment, risk, and returns.

Various business models may be considered in relations to Smart Lampposts:

i. ***Traditional Public / 'City Hall'*** – this is a conventional approach where investment, ownership and accountability for performance rests with the municipality. This model will involve public procurement for purchase of assets, installation, and potentially operations and / or maintenance.

ii. ***Industry Pilots / Research*** – these are fashionable where TRL levels are advancing however risks still remain; or where a particular player seeks market entry. For smart lighting these conditions are less favourable. And though the smart services opportunities are still at less-well-developed TRL the financial value of smart equipment is significantly less than that of lighting assets, so industry players that provide smart services are less likely to invest.

    That said, providers of 'bolt-on' smart equipment may be prepared to invest – this will not however affect a city's ability to upgrade to LED.

    New industries and sectors may be prepared to invest, though not per se by means of research budgets, for instance media or connectivity – where they see the attraction of additional revenue. These tend to be precursor investments to concession models.

iii. ***ESCO*** – energy savings companies have been set up in several cities and regions to share investment, risk, and reward – and indeed reward based on performance incentives. These may involve a combination of: utility, public body, service provider, financial investor. Duties and responsibilities are fashioned to suit. The resulting organisation is a commercial enterprise (though may have public ownership).

iv. ***Concession*** – such business models involve the city offering the operations of a service over a fixed length of time (say 5 – 20 years). In essence a concession is an outsourced contract in which the outsourced provider / consortium will have rights to exploit the service and recoup investment through managing operations most efficiently and / or take (a share of) revenues generated.

    Examples of concessions are emerging where financiers hold overall accountability and contract best available technologies and services; with more open-book accounting to the city. Such models can provide fairer share of revenues and value, respect of public value, mitigate the risk of technical / vendor / provider lock-in, and offer greater flexibilities to cater for change.

v. ***PPP (3P)*** – public private partnerships have been in effect for decades with mixed experience and supporters. Typically, as often with concessions, there

will be an initial investment hurdle that the city seeks to overcome by sharing investment, risk and reward.

vi. ***Community Ownership*** – this may take the form of PPPP (4P) models, or straight ownership of smart lampposts. The latter is relatively new and may involve a social community or more likely retailer taking ownership of the asset, and having greater control of services that can be generated from it. For instance, a retailer may elect to combine light with eV charging or parking management, and see returns (financial or customer service, by offering parking for customers) as a result.

Distinction should be made between '*financing*' – the investment cycle, versus '*funding*' – the returns stream, in business model discussions.

Considerations that cities should evaluate in considering business model selection include:

- Financial strength – of the city budget, and provider
- Regulatory, policy & other contextual settings
- Policies and norms for asset ownership (public/private)
- Policy and practices on asset operation and maintenance
- Flexibility to cater for change
- Risk and risk propensity
- Ambitions to control the market
- Policy on revenue generation
- Technical, commercial, business management and other competencies

5.10.2 Sources of Finance

Not unrelated to the selection of business models are considerations on sources of finance. These may include any of the following:

Public

- Borough capex / opex (revenue) / other budgets
- Government grants
- Sovereign funds / loans
- EC H2020 project grants
- EU project development grants / assistance (eg EIB)
- EU Structural and Investment Funds
- Institutional grants and loans

Private

- Commercial Market
- Pension funds / Insurance
- Philanthropic
- Industry

Blended

- Mixed funding sources

Loan terms are a function of risk and opportunity. The ability of a city to reduce risk through quality management of implementation; and demonstrate returns through

quality measurement. And the ability of a city to bring scale to the investment market are all factors that cities should address.

5.11 Procurement / Commissioning Templates

…in development…

This section will capture examples of procurement templates to refer / synthesize / include in annex.

5.12 Operational Savings Metrics / KPIs

Measurement of savings will be vital to demonstrate decision makers and society that the smart lamppost is of value. We have included below the text from ShC deliverable 8.1 of "Common monitoring and evaluation framework (CMEF)" which is the master document for the programme. As such this may help stimulate measurement activities in other cities. Consideration should also be given to other measurement systems: e.g. locally applied, or CITYKeys.

It should be noted that the ShC Lamppost evaluation is based **only** on lighting benefits. Indicators for benefits from smart services and features will need to be developed (and as they do from Sharing Cities work or other sources the intent is that this document will be updated). These may include financial gains (cost saving; capital avoidance; revenue); and non-financial benefits – or benefits that are most meaningful not converted to finance (safety; perceptions; health; convenience; time; etc), and these will be dependent on the specific smart feature(s) that are implemented.

The CMEF text follows…

3.4 Lamppost

Aside the anticipated installation of more energy efficient LED lighting, as discussed in Section 2 (Table 2.4.2), streetlamps can accommodate several functions to contribute towards an improved urban efficiency and performance. Lampposts are relatively simple city assets that all too often are purchased at an individual city level in relatively low volumes (to often bespoke specifications). An integrated function lamppost is by nature a highly replicable and scalable solution that can have a modular form to accommodate functions depending on local needs. Therefore, the impacts of a lamppost are highly open-ended as they are module/ function based. The evaluation targets proposed, focus on the fundamental functions of a lamppost (i.e. LED lighting and light dimming) and its utilization level as street furniture. Evaluation targets can be further extended to accommodate targets for individual functions. For example, for eV charging capability of lamp posts, the evaluation targets presented in section 3.3 on eV charging can be used.

3.4.1 Technical performance

The technical performance evaluation targets of lamp posts considered focus on lighting and adaptive light control (dimming).

- *How much energy is consumed for lighting?*
- *Is lighting provided sufficient?*
- *How accurate are the data collected?*
- *How many modular functions are accommodated?*
- *How much maintenance is required?*

3.4.2 Impacts on attitudes and behaviours

Lamp posts LED lighting and light dimming are anticipated to have an impact on attitudes and behaviours of citizens, operators and stakeholders. The evaluation targets considered monitor:
For residents:

- *How satisfied are residents'?*
- *How satisfied are visitors?*
- *Does lighting effect route choice in walk trips?*
- *Does lighting effect route choice in vehicle trips?*

For operators:

- *Does their perception of system functionality change?*
- *Does their perception of system control change?*

For local policy makers/ stakeholders:

- *Does their willingness to install new smart lamp posts change?*

3.4.3 Wider systemic impact
The lamp posts demonstrators wide systemic impact evaluation targets require to focus on quantifiable impacts, while at the same time be compatible/ comparable with evaluation targets of other demonstrators. In that context, the wider systemic impact evaluation targets considered are:
Lamp post specific:

- *Is road safety influenced?*
- *Does local criminality change?*

City-wide:

- *Does lighting energy efficiency change?*

3.4.4 Institutional & business consequences
Lamp post demonstrators' institutional and business consequences are limited when only LED lighting and light dimming functions are considered. Therefore, the following evaluation targets for lamp posts act primarily as a feedback loop for future demonstrators:

- *How successful have the procurement mechanisms been?*
- *How has the productivity of the affected area changed?*

3.4.5 Economic and social implications
With respect to lamp posts demonstrators, economic and social implications include the indirect economic and social effects on the local population. Therefore, the following evaluation targets are considered:

- *Does street lighting upkeep cost change?*
- *Does local land value change?*
- *Does neighbourhood liveability change?*
- *Does the demonstrator encourage social inclusion?*

Evaluation target	Measurable indicator	Unit	Data collection method
How much energy is consumed for lighting?	Energy use		Lamppost logger data
Is lighting provided sufficient?	Luminescence	Lux (lm/m^2)	Light sensor
How accurate are the data collected?			
How many modular functionalities are accommodated?	Amount of functions		Lamppost logger data
How much maintenance is required?	Frequency of minor repair	Time (or km) between repairs	Operator survey
	Frequency of major repair	Time (or km) between repairs	Operator survey
How satisfied are residents?		Grade (1-5)	User survey
How satisfied are visitors?		Grade (1-5)	User survey
Does lighting effect route choice in walk trips?	Path directness i.e. path distance / straight line distance	%	User survey
Does lighting effect route choice in vehicle trips?	Path directness i.e. path distance / straight line distance	%	User survey
Does operator perception of system functionality change?		Grade (1-5)	Operator survey
Does operator perception of system control change?		Grade (1-5)	Operator survey
Does stakeholder willingness to install new smart lamp posts change?		Grade (1-5)	Stakeholder survey
Is road safety influenced?	Safety incidents		Municipality data
Does local criminality change?	Criminal incidents		Municipality data
Does lighting energy efficiency change?	Energy use per luminance provided	kWh/Lux	Lamppost logger data

To accurately quantify the effects of lamppost demonstrators it is required to capture or monitor (if applicable) several influencing factors that can potentially create biases in the data. The influencing factors for lampposts evaluation targets and measurable indicators include:

- *Safety incident severity,*
- *Criminal incident severity, and*
- *Local weather conditions, including:*
  - *Temperature,*
  - *Humidity,*
  - *Wind speed, and*
  - *Precipitation,*

Summarizing the data collection methods for all evaluation targets and measurable indicators, the following data are required per mobility demonstrator include:

- *Lamppost logger data, including*
  - *Energy used, and*
  - *Active modules*

An example of how the above aspects of the toolkit which make up the Financing and Procurement section of the toolkit has been applied to date in the Sharing Cities programme and lighthouse cities is given below.

Applying the toolkit in Sharing Cities: Finance & Procurement

All cities have begun to consider the financing of smart lamp post solutions. Through trailing both light and non-light use cases we can begin to build models of investment made and the costs associated with each use case, which can be taken forward as the basis of business cases for roll out of measures within the city, and in other cities across Europe. The first step, particularly in the case of lighting UCs (LED and CMS) has been identifying how much the asset currently costs; and to establish ways of monitoring savings generated from applying the lighting UCs.

For example, Lisbon network for public light (PL) comprises around 67.000 light points, with 16.6 MW contracted power and an annual consumption of 55 GWh. For the current average cost of electricity for street lighting (SL), 0.14 €/kWh (including VAT), the 2016 electricity was around 6 M€. In Greenwich, data gathered through the trials of LED and CMS replacement will help inform the business case for its roll out.

5.13 Benchmarking / Bench-Learning

Benchmarking is the process of comparing the performance of a system, based on specific measurements, to a set target and / or against others. This process can lead to helpful habits which support performance improvement. In a complex systems-of-systems environment – like cities: (i) it is however very hard to ensure any form of consistent context between cities; (ii) it is hard to know where actions from another system or process are having an effect, and (iii) 'public value' is a much more complex thing to measure.

As a result, benchmarking risks leading to consumption of much staff resource worrying about ensuring the right numbers are reported to senior officers, politicians and press. It can be the pursuit of perfection where such measurement is unattainable.

Bench-learning is a term that has been in used in a European public sector context for a decade or so. It has proven to be highly effective. Bench-learning is distinctive in that it:

- Puts more attention to developing a *framework* for measurement
- Ensures greater understanding of the logic and process of measurement, rather than the number that results from it; so, it helps in the understanding of measurement of inputs, process, outputs, outcomes, and impact
- Seeks to provide a basis to capture the context, the setting, within which measurement is taking place (so that parties can have a structured means to compare context – useful of itself)
- Focuses on *accuracy*, over *precision* (and thus perhaps the accurate capture of a number of factors, as opposed to the myopic focus on the 'one number' benchmark)
- Ensures that time is spent amongst those involved to compare practices, dialogue, test – adjust – monitor, and thus learn and improve

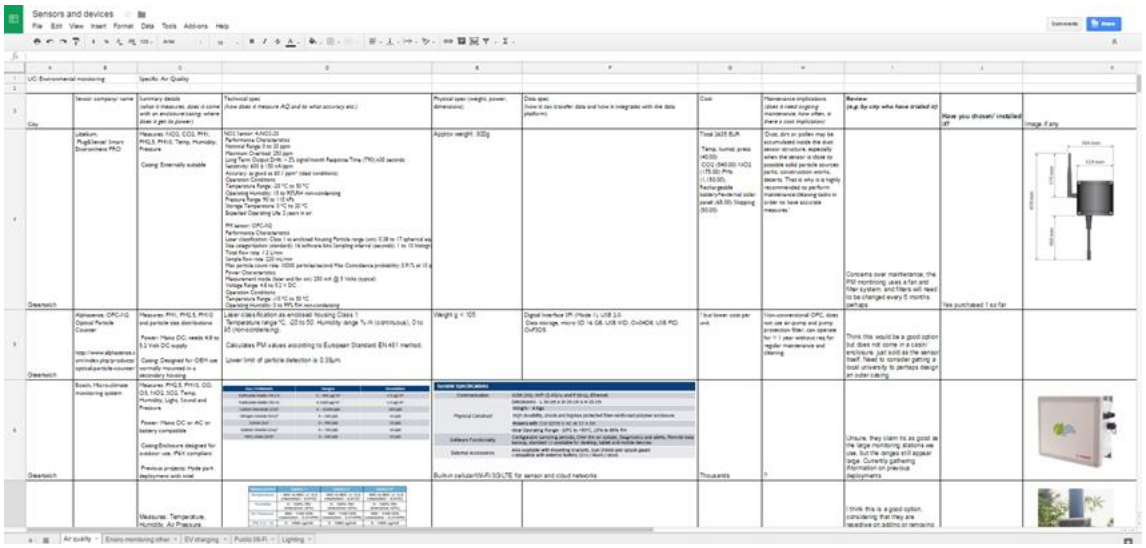
In the context of cities, which are complex systems; addressing the complexity of 'public value', with constrained resources, bench-learning can be more fruitful as a means to sustaining improvement.

The EIP-SCC, fed by a growing number of SCC01 Lighthouse programmes, offers an effective route to bench-learning. National (& international) city associations can also play a useful role in his respect.

An example of how the 'Operation, Monitoring and Sharing' section of the toolkit has been applied to date in the Sharing Cities programme and lighthouse cities is given below.

Applying the toolkit in Sharing Cities: Operation, Monitoring & Sharing Learning

Each city in the sharing cities programme is yet to fully apply this last stage of the toolkit, however the ethos of sharing is very much at the heart of the programme, and the three cities have regularly shared experiences and learning so as to advance the UCs as far as possible. This has been particularly valuable in the case of the sharing of details and experiences with specific technology and devices that can be used to implement the use cases; for example small scale air quality monitoring devices. The three cities have set up a shared directory where they log details of devices they are exploring, this enables the programme as a whole to consider a wider range of technology and whether it enables an effective UC to the city; but will also provide cities outside the programme with a record of what has already been considered. A snapshot of this directory is shown below.



6 Technical Specifications "Bill of Materials"

This section provides mode detail on the various components of the Humble Lamppost.

6.1 Smart Lighting Components

The tables below provide further information about the various major components of the smart lamppost system. This is not intended to be a detailed engineering specification – that can be found in the various relevant standards. Rather, it is intended to provide an overview of the elements, that may prove helpful for stakeholders involved in smart lamppost projects.

1. Lighting Column / Pole / Lamppost	
Function	*Provide a sturdy elevated stanchion to locate the luminaire / light*
Requirements	*Typically manufactured from steel, cast iron, concrete, or wood.* *Heights vary incl. 4, 6, 8m dependent on location.* *Powered (preferably 24x7 for smart services).* *Traditionally used to hang signs, flowers, and the like.* *Field life in decades.*
Relevant Standards	*Old vintage standards (ref annex for national publications)*
Typical Asset Owner	*Municipality (mostly); Utility;*
TRL Level	*9*
Market	*Mature, fragmented, local, low margin*
Cost range	*Traditional: €hundreds; Multi-functional smart: €thousands*
Areas of uncertainty / challenges	*75% are >25 yrs old. Maintenance costs can be high, and condition poor.* *Wind loading a concern if new fixtures added to old columns.*
User insights	*Multiple variations exist from very basic to highly decorative. Historic areas frequently have policies to retain decorative columns.* *There are around 6-10 people / street light in a city (useful for rough calculations)*
Example Pictures	



Traditional lighting columns *of various materials and construction*



Street signage



Hanging baskets



Galvanised steel lighting poles

2. Luminaire / Light	
Function	*Improve safety for driving, walking, cycling.*
Requirements	*The vast majority of street lamps are high or low pressure sodium HPS/LPS in production since the mid 90s; they produce 'warm' yellow spectrum light.* *Modern lights are typically LED with multiple lamps that enable greater light control. LED is more targeted / directed with less stray light pollution and bright white light.*
Relevant Standards	
Typical Asset Owner	*Municipality (mostly); Utility*
TRL Level	*Sodium: 9. LED: 7-9*
Market	*Mature, fragmented, local, low margin*
Cost range	*LED*
Areas of uncertainty / challenges	*Health effects of LED lights still under study – white light can disturb circadian rhythms. However whiter light delivers safer environment for driving etc.*
User insights	*LED dimming to 50+% of power is typically not noticeable to the public, though saves considerable additional energy.*
Example Pictures	



Contrasting sodium and LED lighting



Traditional sodium luminaire



***Examples of** LED luminaires*

3. 'Smart' Fittings	
Function	*Exploit the lamppost asset for other 'smart' services*

Requirements	*Generally 24x7 power supply (unless 12hr services and/or battery back-up and/or PV power). Refer to various use cases to determine the nature of the ancillary equipment required to provide additional services. Potential emerging future requirements for high volumes of IoT data; potential for infra-to-vehicle (& V2I) communications.*
Relevant Standards	*DIN PAS; plus multiple potential detailed specs (country specific). Many refresh specs will be required as technologies and opportunities emerge.*
Typical Asset Owner	*May span public, private (& potentially community) sectors. 'City Hall' should in principle 'convene' the desired outcomes from these assets (unless they are within a bounded privately owned area – eg University campus; airport)*
TRL Level	*Variable from 4 to 8-9. Fairly technical dynamic territory. Though the logical design may stabilise the physical realisations are likely to go through considerable change.*
Market	*Typically more regional / local, with increasing number of SMEs*
Cost range	*Highly dependent on use case being served. Multi-functional poles that (in principle) provide for several functions are in the €5-10k range. Individual sensors and equipment are in the hundreds and low thousands range.*
Areas of uncertainty / challenges	*Determining the density of services (to suit community needs, or technical capabilities). Selecting the right combinations of stable technologies to take action (cf just wait, which erodes value from potential energy savings). Service contracts (which can involve considerable length) in some cities may inhibit / dis-incentivise the addition of smart services where they are perceived to add risk, costs, and resource burden outside of the core contact / contractor expertise.*
User insights	*Multiple and early days – to be further captured. Potential wind loading issues from additional equipment. Future-proofing decision making.*
Example Pictures	







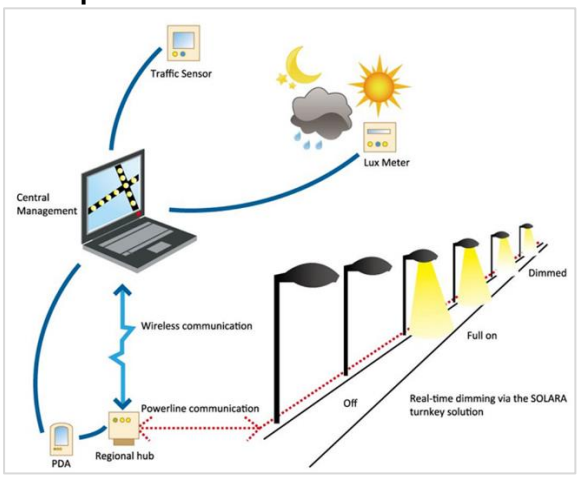
PV / Wind power *Connectivity* *CCTV* *eVehicle Charge point*

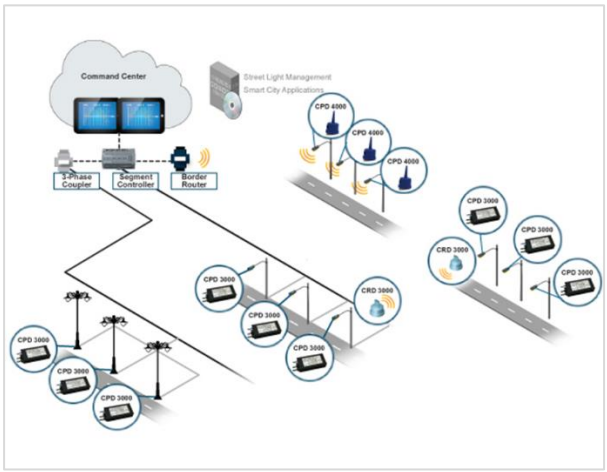
4. Electrical Systems

Function	*Provide power to individual or groups of street lights. And now also to the ancillary smart service equipment (the need often for 24x7 power)*
Requirements	*Low voltage power. For smart services: 24hr power service*
Relevant Standards	*Typically, country specific standards. Ref annex for initial list, and National Standards Body for detail.*
Typical Asset Owner	*Power Utility (potentially city-owned)*
TRL Level	*9*
Market	*Very mature. Typically international and national suppliers of equipment; and national and local operators.*
Cost range	*Variable, dependent on equipment.*
Areas of uncertainty / challenges	*24hr supply for 'smart' features.*
User insights	*Communications with utility; current & future billing systems can prove challenging to shift from conventional lighting services to smart features (may involve considerable negotiations). Power usage for all except eVeh charge points is modest.*
Example Pictures	 In-pole light control  Street power cabinet

5. Control Systems	
Function	*Provide system wide lighting and smart function control. Often referred to as Central Management System (CMS)*
Requirements	*Basic on/off function; 'smart' lighting involves greater levels of control for ambient conditions; local specific circumstances (e.g. accident that requires full light levels). "Trimming and Dimming" features are enabled to provide light (group) specific adjustments. The line between CMS providers and broader 'urban platform' providers is greying.*
Relevant Standards	*Several international IEC/ITU standards and national standards. Reference also the EC EIP-SCC Urban Platform suite of guidance for broader 'smart' urban platform and data management considerations.*
Typical Asset Owner	*Lighting Service Operator (municipality, utility, or contracted provider); and potentially city technology provider (for urban data platform services)*
TRL Level	*7-9*
Market	*Served by lighting industry OEMs (traditional); city service providers and multi-utility companies. Growingly also by national & local SMEs*
Cost range	*€X million, or lower for smaller cities / simpler systems*
Areas of	*Change from lighting specific (potential lock-in) CMS system to city-*

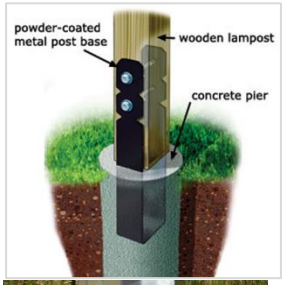
uncertainty / challenges	*wide urban operating platform.*
User insights	*Industry Associations and pseudo-standards organisations may provide additional insights (eg TALQ)*
Example Pictures	





CMS overview configurations

6. Civil Engineering	
Function	*Provide secure fixture for (smart) lamppost.*
Requirements	*Affordable secure base (as it will likely be in place for 30+ years)*
Relevant Standards	*Typically national civil engineering norms*
Typical Asset Owner	*City / Utility*
TRL Level	*9*
Market	*Fragmented local*
Cost range	*€X thousand installed*
Areas of uncertainty / challenges	*Few.* *Allowance for additional (horizontal wind and other) loading*
User insights	
Example Pictures	







Wood column footing foundation with cable ducts Installed pole …(awaiting pavement)

7 Lessons Learned: Critical Aspects and Key Recommendations

8 Conclusions

This section provides concise conclusions for this publication.

"The Devil so often is found in the detail". What on the face of it might appear to be a relatively straightforward solution for a city – to upgrade its street lighting assets to provide additional smart services – can so often get stuck in technical, organisational, operational, financial and economic, and at times regulatory challenges.

That said, smart lampposts are an important and very visible quick win for a city, that clearly demonstrates a move to embrace 'smart city' opportunities, and at the same time offers a handsome return on investment.

Successful implementation requires the involvement of a stakeholder group that extends far beyond traditional lighting engineers. And it is particularly because of this that this publication has been produced – to aid cities of all sizes to move more swiftly to adopt smart lighting solutions. And it is focused on *cities* particularly – as progress starts best from the demand side. Supply and financing will follow if cities move with confidence and more consistency.

The opportunities for common solutions applied, at scale; with common approaches; potentially common (logical) designs; and possibly shared procurements ('demand aggregation'), are very real tangible and timely.

In order to secure smooth evaluation, decision making, financing, technical implementation – and from there, ongoing operational performance and sharing of experience / benchmarking – there are some very practical steps that can be followed. These have been captured herein.

The proposed approach and tools are considered to be a 'good start' from which additional learning will support ongoing improvement. As such, the other SCC01 'Lighthouse' programmes and the EIP-SCC Humble Lamppost initiative, supported in time by the open and active participation of the supply and financing communities, are important companions, and ongoing actively-managed collaboration will take place.

This publication should be read in conjunction with various other guidance – some within Sharing Cities (e.g. Use Cases); notably the new DIN (PAS) standard; existing national standards; and emerging industry association and industry guidance.

Moving forward, this document is planned to be offered for publication via the standards bodies (national and/or international) to provide trusted and managed guidance to help cities accelerate the adoption of smart lampposts. And to support industry in ensuring they respond to more clearly defined needs, and can thus deliver affordable solutions that engender confidence in cities.

9 ANNEX

9.1 Humble Lamppost Maturity Assessment Matrix

The 'Development Assessment Matrix', included in Section 1.1, is shown here in readable format.

Humble Lamppost: 5-Step City Development Assessment		Maturity Assessment Rating				
Element	**Goal** (what is a notional ideal leading practice future state?)	**On-Ramp 1: "Traditional"**	**On-Ramp 2: "Basic LED Upgrade"**	**On-Ramp 3: Basic Smart w/New BizM**	**On-Ramp 4: Adv'd Smart/New BizM**	**On-Ramp 5 Leading Practice**
Lamppost / Luminaire Asset	**Common logical architecture design at EU (EIP) level, supported by EU / International Standards Orgtns. Flexible 'component-based' solution that cities can select as appropriate (inter-operability). 'Future proof' design that industry apply innovation to. Affordable, and template based / easy-to-procure packaged proposition.**	Aging physical pole no / limited ability to take additional attachments. High consumption luminaires. High down-time; hi maintenance.	Current	LED luminaire selected to suit location.	Desired	Component based smart lamppost, future-proof design for smart upgrades. Dynamic interoperability with city systems.
Societal view	Attractive lamppost design with pleasantly lit communities, with flexibility in operation (e.g. dimming; artistic lighting etc). Safe place (e.g. 'push to talk'; brighten when danger). Optional Smart Services enabled (e.g. eV charge; WiFi; Air..). Respect of personal data privacy (e.g. geofencing for retail offers).	Ugly old poles. Poor quality lighting. Sense of danger in certain areas. Multiple call centre complaints		Light levels measure background building / retail light levels to set lamppost light.	LED luminaire selected to suit location (RGBA-W/Tri etc) - mood /façade lighting	Community (residents AND businesses) budgeting and lighting / smart services. SLA reporting.
Built Env't interface	Integrated bldng/place based lighting, with movement controls, safety and light level sensitivity. Auto-fault reporting. Smart services commissioned by local community groups. Façade and artistic / mood lighting where appropriate. Auto integration between 'place' and lampposts (eg power storage).	Badly placed street lights irritating residents (hi / low intensity areas). No integrtn between place & l'posts. Poles detract.		CMS (central mgmt system) for lighting level controls at street / area level.		Integrated building-place / lamppost lighting in design and operation. Mood / safety lighting.
Energy Mgmt System Interface	"*Energy Efficiency*": PV to power L/post. RE-FIT model to fund L/post. Monitoring. "*Store & Control*": Monitoring power consumption (accurate). CO2 emmossion monitoring. Tarrif incentive. Storage (battery). Demand response asset. eV power; & eV to light. Licence life business model, revenue stream. Mobile phone charge. Store/Supply/Generate/DSR. Off-grid l'post. Micro-grid.	Street level energy measurement (often less).	No CMS. Measureemnt of energy consumption at area based level.			Auto integration and light / energy optimisation. PV/Alt energy lihting systems.
Mobility interface	Mobility services; on/off & dimming as function of traffic flow. eSignage option. eV charge; eBike interface. WiFi mesh to better access eV charge network. Link with connected/assisted driving (infra-car etc).	Lampposts and lighting levels poorly integrat3ed with transpot system, with resulting risks.			eBike/Vehicle charge points with locatn-driver apps. Lamp to ITS interface.	Incentive / demand response apps to condition behaviours and optimise flows thru time-of-day
Data Platform	Defined sensor interfaces (gateways). Clearly selected (open) protocols. Defined interoperability layer (sharing). Identifies lamppost and lighting status. Identified use cases for data.	No systems support. Lamps set to mechanical timers, no automation.		CMS integration with city platform.	Lighting / Lamppost-enabled apps. Data strategy (eg geofencing) to respect privacy & enable services.	Due proection of personal and asset data, yet innovative data services, with city / personal choice
Business Models Financing & Funding	Demand aggregation locally; networked internationally to enable best price / innovation mix; and support local SMEs. Business Models that incentivise aggregation; support x-silo delivery and liberate value. Multiple proof points for financial social ecological value.	Silo annual budgets that inhibit investment / long-term decisions. Constricting procurement practices.	Price-based LED upgrade (little demand aggregation).			Concession models. City-region commissioning. City-data monetisation for city/community.
Evaluation	KPIs: e-Charge (# charges per day; KW/day) Nos of useful 'push-to-talk' uses. KWhpv/KWh used. Nos Failures. Nos queries to answer	Outage complaints inefficiently handled. No/basic energy KPIs. No effective targets. Silo'd Ops costs.	Financial measures to demonstrate energy cost savings.	Public light levels set on top of background light levels to reduce public energy consumption.		Triple bottom line metrics, linked to commercial-retail monetisation / residential bonds
OTHER	Public Regulatory & Policy conditions that support swift commissioning / procurement; local business enablement and innovation.	Regulatory barriers. Constraining local policies. 'Silo' thinking and behaviours (almost encouraged).				Liberal regulatoiry environment that supports innovation. Open policies.

9.2 Asset Landscaping Example (Lisbon)

The Lisbon network for public illumination comprises around 67,000 light points, with 16.6 MW contracted power and an annual consumption of 55 GWh. With the average cost of electricity for street lighting, 0.14 €/kWh (incl. VAT), 2016 electricity consumption was ~6 M€.

Presently, around 2,700 lampposts have LED technology (4%) but most without a CMS system. Several projects for streets and squares rehabilitation are starting, forecasting that at the end of 2017 around 6,000 lampposts will have LED technology. Lisboa E-Nova is working with the Municipality to maximize the number of lampposts with CMS system and smart features.

The Municipality is also considering another project, to convert all the lampposts, including post intervention as a significant percentage requires intervention. The business model for this large action is not already defined, but EPC funds are considered.

There are several types of lampposts in the city of Lisbon's demonstration area. These are described in the table below.

Table 5 Types of lamp posts and quantities

Type	Description	Nos	Image	Type	Description	Nos	Image
Normal lamp post	Light post with height equal or higher than 4 meters	1,741		**olumn amp post ith height 4m**	Normally used to light pedestrian areas, for example sidewalks and parks	1,450	
Composite	Used to light both sides of the road	268		**echnical ght quipment**	These types of lampposts are used when the infrastructure doesn't allow other types.	536	
Street wall lamp	Used to light the road in urban areas and historical areas	5778		**uste**		115	

The location and state of these lampposts are well documented. The location of each one of the 9,888 lampposts in the SC demonstration area is represented in the next figure.

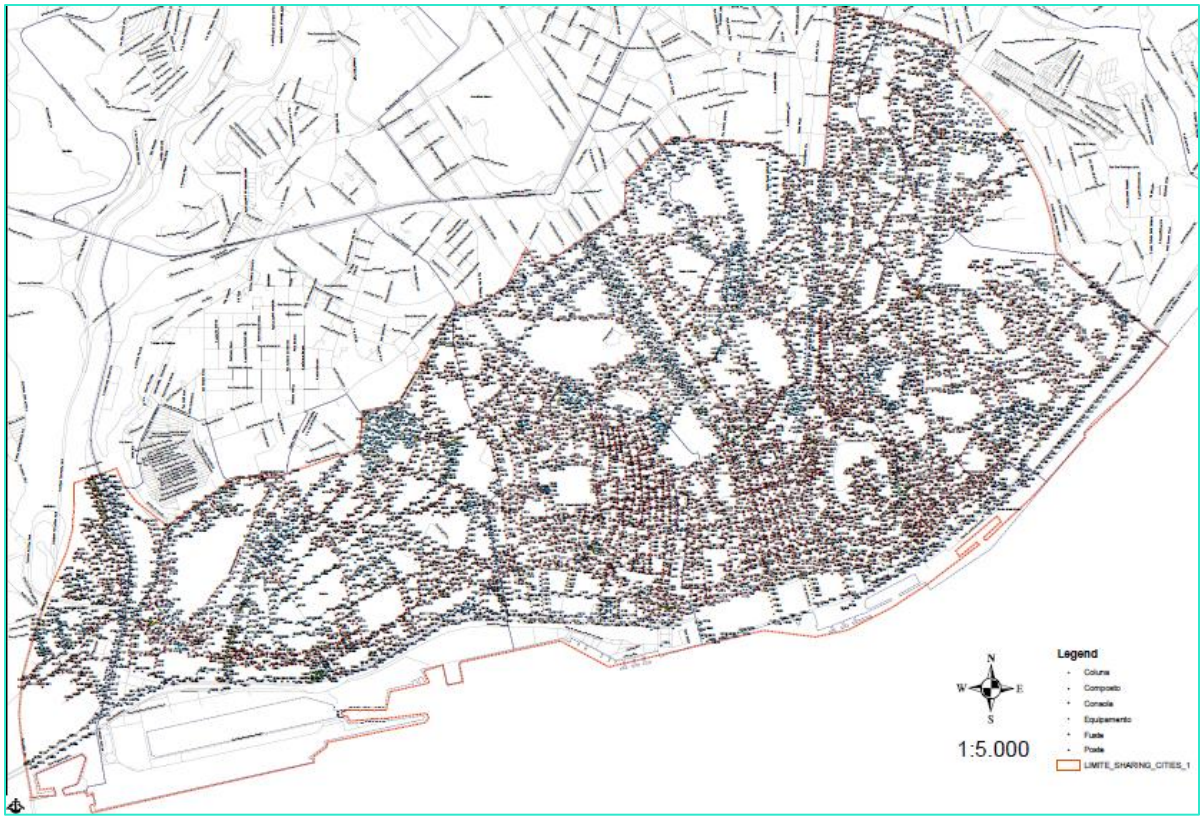


Figure 18 Lamp posts in the Lisbon SC demonstration area

New developments are currently being made in the demonstration area regarding LED and CMS conversion. The figure below shows the areas where this conversion is planned/taking place.



Figure 19 Area where LED and CMS conversion is taking place

In total, xxx lamp posts are going to be converted to LED, but not all of these conversions are going be coupled with a CMS.

9.3 Use Case Capture Template

Use cases generally follow a similar format. One of the several Sharing Cities examples is shown below (the full set are available in D3.11). In addition to this descriptive use case, Sharing Cities has developed an XL Data Capture proforma which may help cities in data management plans. DIN Use Cases may also be consulted (through purchase of the referenced PAS).

Use Case : HL #2	**Additional Energy Savings / Optimisation**
Description (narrative)	As a city manager, I wish to increase the level of energy savings beyond just LED lamp replacement – by trimming and dimming the lights to suit environmental and other circumstances
Functions	Sunset / sunrise times are incorporated in the system Ambient light conditions are monitored Footfall and traffic volumes may be monitored to reduce light when volumes are low Political / societal opinion and feedback is captured and built into lighting controls Reporting on a regular basis for issues, and for energy consumption (over/below norms) to ensure value is maximised and balanced against other factors.
Pre-condition	
User Insight/Need the Use Case Responds To	Assuming citizen user is referred to here, there is little articulated need specific to this use case. Indeed it is the counter issue (ie insufficient light) that is the resulting down-side risk that would engage the user.
Actors involved (stakeholders)	City Council; Transport Authority; DSO; User (Acceptance); Businesses; Energy Agencies; Meteorological agencies (advanced warning); Data Consumers
Incentive for Citizen	Limited. Savings generated are unlikely to be visible in eg council taxes.
Description (narrative)	City council environment / energy officers wish to make energy cost savings (and to an extent GHG reductions) through optimising street light intensity levels depending on local weather patterns and other factors (passage of people and cars) through smart street light controlling, whilst still providing appropriately attractive and safe environment for society.
Functions	The streetlights are installed with LED luminaires and a CMS (control) system that has the capability to dim individual lamps (or groups) based on optimising ambient light conditions with required set lighting levels, adjusted to cater for low footfall or low traffic flows (eg night time). This results in additional savings over and above just upgrade to LED. Assumption is LED is in place, however further additional benefits are available through: • Dimming dependent on traffic flow – area specific • Trimming due to ambient / environmental light levels (eg temporary rain / storm) • Dimming through high local (eg retail) lighting levels Excluded as these are not savings; they're additional energy use • Trimming through safety and public movement
Pre-condition	
Policy	Policy focus on energy savings, and preferably support for smart lighting
Legal & Regulation	No specific issues envisaged for this use case
People	No significant input required. Support for (lamp) outage reporting a benefit (not for energy, more for safety)

Operational	Clarity of role of city and / or service provider; and agreed collaboration
Process	Existing operational SOPs are in place as a basis to work from
Data	Energy consumption data are preferably lamp levels (if not then in small clusters). Weather (light levels) data, preferably thru local ambient light sensing.
Technology	• CMS system in place • Grouped controllers • Sensors for ambient lighting levels (at beyond individual lamp level) • Movement tracking (eg cars; people)
Assets	Modern lighting poles; multi-bulb LED luminaires; CMS system; Sensors
Performance / Criticality	Low risk use case; deals with system optimisation
WP3 (PLACE) Measures	
	Smart lampposts *Sustainable Energy Management Systems*
WP4 (DATA PLATFORM) Requirements	
Sensors/Devices	Ambient lighting level sensors Vehicle and pedestrian movement sensing or better, flow monitoring
Applications	Typically applications available as part of existing Industry propositions
Data Acquisition	Near real time data for light-level optimisation; real time for vehicle/people flows
Data Usage	
Data Flow	Light level sensors feed CMS system, which then sends control signal to lamp. Movement sensors feed CMS system (unless sensor-to-lamp group controller?) to control LEDs in specific lamps.
Data Volume & Frequency	Info packet is small. Frequency for ambient light optimisation is in principle twice per day over two periods of around 1 hour. For movement optimisation it Will be much more frequent during the dark hours.
Protocols	Standard messaging/communication protocols anticipated (detail not known)
Datasets	Energy consumption data; light level data (sensed) and publicly provided (dawn/dusk times); inclement weather data (storms); traffic/pedestrian flow data (higher volume) Type of information handled by each device, e.g., traffic data, environmental data. Please reference existing standard data models, ontologies, schemas etc. when applicable
Data Analytics	Energy consumption trending data analytics and reporting.
WP2 (PEOPLE) Requirements	
User profile	User is city official
Incentivisation mechanism	No citizen incentive mechanism anticipated
User interface	None
Citizen Engagement	Only for advising (and as part of larger smart lighting / services package)
Business Engagement	Optimisation in high street to exploit existing business lighting to dim public lighting may require some engagement with businesses.
Realization	
Main responsible partners	• Lisbon: EDP; Lisbon Council; • London: RBG; GLA; TfL • Milan: A2A; Milan Council; • Burgas: Council;

Use Case ready for PILOT (year 1 operation)	Yes, if one of cities accelerates smart lighting implementation – checking RBG status; potentially in Milan using existing LED upgrade?
Impact of Use Case	Financial savings; modest GHG reduction
Contributing partners	WP 3.4, 4, 8
Priority	Medium / High

9.4 Use Case Selection Example (Lisbon)

USE CASE SELECTION

Lisbon's Use Case selection is focused on 4 different aspects the city considers to be important to address, particularly as they raise concerns for residents:

- L1 - Light: LED conversion has been an aspect that is being implemented by the municipality. Plans are being carried to couple this conversion with CMS to take full advantage of these conversions in 2018.
- NL4 – WiFi: this use case will be applied to squares were the Municipality implemented the free WiFi systems.
- NL6 - Environmental sensing: This use case was established from the beginning, regarding the need to implement an environmental monitoring system in order to address air quality issues.
- NL6 - Noise measurement:
    - In WiFi zones: the sensor is based on a multi-microphone system, allowing capturing several sound streams, that allows the sound to be reproduced with directionality and allows the identification of the direction of the maximum registered sound. This system will allow a deep emersion in what we can call the "sound of Lisbon".
    - In LoRa zones: the sensor will only capture the noise level.
- NL12- Movement: this use case is oriented to vehicle and people counting, indicative of the city's traffic.
- NL2 - Sensor integration: this use case is aimed to identify cross relations between the row data captured in the previous use cases, that can inform traffic management teams improving the city' traffic control.

COMMUNICATIONS

The city will have two different communications solutions for the sensing data:

a) Wifi in special squares, where the municipality develop rehabilitation projects
b) LoRa for the entire Sharing Cities intervention area

MAPPING OF THE UC TO THE DEMONSTRATOR AREA

This intended in 2017, to study how frequently the devices should be installed on the lamppost to be able to cover the demonstration area.

9.5 Market Research & Analysis Example (Lisbon)

MARKET RESEARCH & ANALYSIS

A market research and respective engagement was made to seize information on the available technology (sensor, devices, and communication protocols) that could better fit in the UCs listed above.

- Air quality: research into devices that can easily be installed on lamp posts, offer a solid way to disseminate the data acquired by the device and that its appearance is going to fit on the environment where is installed. Noticed that the market was not fully developed yet and was changing fast, with a lot of different options regarding the quality of the sensors offered. Another issue found on this market was the fact that the quality highly affected the price per unit of these devices, and being the aim to produce meaningful data, the prices for the desired outcome would be high.
- LED conversion- The city is underway with the conversion of the existing luminaries to LED during 2017. It's predicted in 2018 to couple this LED conversion with CMS.

Energy supply is an important topic in Lisbon, as the large majority of lamp posts just receive electricity during the night. That obliges to consider a power supply storage device which can be charged by the lamp post during the night or by small PV panels during the day. This affects the type of sensors and data volumes as the battery capacity is limited.

However, in the lamp post already converted to LED with CMS would be possible to supply 24h continuous voltage, overcoming these challenges.

DEVELOPMENT OF SPEC (see also notes below for each)

After the initial market research, it made sense to divide the development in 3 parts:

1. ***Enclosure***: currently working with the local FabLab to define a common city-wide enclosure for all sensors, allowing modularity.
2. ***Sensors Acquisition***: The 1st phase includes the acquisition of 100 sensors for an initial deployment. The 2nd phase consists on the acquisition of 400 more sensors, depending on the results of the 1st phase and the plans include market participation.
3. ***LoRa Network***: Definition of the antennas and gateway locations with the help of local companies.

FORWARD PLANS:

- LoRa network coverage.
- Deployment of the first 100 sensors to assess how they operate.
- Connect the data acquired with the USP.
- Convert more lamp posts to LED lighting.

NOTES

1. Enclosure

Representing the outer appearance of the system, the enclosure, raises a concern of aesthetics that can't be overlooked. Its appearance can determine whether the integrity of the system is preserved or not, meaning that if its appearance is too flashy it can be vulnerable to theft or other types of vandalism. Other aspect that is related to the enclosure is the maintenance of a system, for example, if we want to add functionality (e.g. installation of a new sensor) to the system we need to replace it as a whole. But if the design of the enclosure allows modularity, allowing adding or removing modules as needed, when a scenario like the one described before happens the system will adapt without having to be replaced. This would open a new perspective regarding the maintenance of the system.

2. Sensor Acquisition

The acquisition of the sensors is going to happen in 2 phases. The 1st phase will consist in the purchase 100 sensors to deploy on the city and it will be used as an experience to assess the operation of these sensors. The 2nd phase will also consist in the purchase of 400 more sensors with the objective of improving the system previously deployed, based on the flaws showed on the time of operation.

The sensors intended to acquire are going to be consisted of air quality sensors and traffic management sensors.

This sensor system is intended to run 24 x 7, being a possible solution the use of a battery allowing the system to operate continuously. A discussion on the type of energy supply to use is still ongoing, having yet to find the solution that best fits in our specific case.

To have a better inside and understanding of the sensor world, a small sensor system was developed at Lisboa-Enova, which can be seen in the figure below.

ADICIONAR FOTO DO SISTEMA

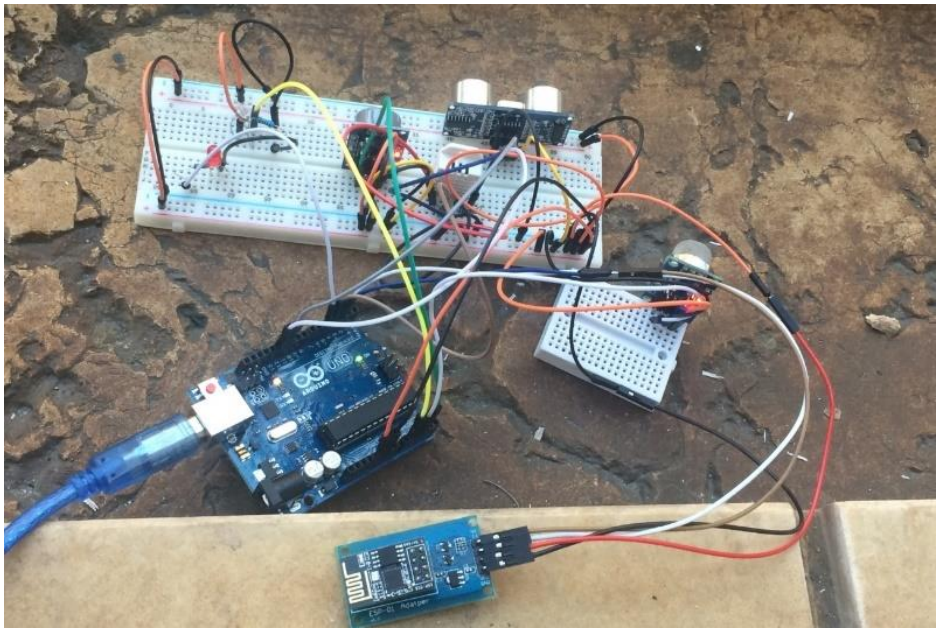


Figure 20 Small sensor system developed

The data acquired by these sensor system is sent to an online platform, using a WiFi module available in the system which is connected to the agency network. This data is available in the platform and as shown in the figure below, this data is presented in a graphic form.

ADD A SITE PRINTSCREEN

This system is Arduino based, powered by the electrical grid and is composed by several sensors:

- *Temperature and Humidity sensor*

This sensor is responsible for the data related to the **Temperature** and **Humidity** in the atmosphere, and having these 2 values available we are able to calculate the heat index or the "apparent temperature".

- *Ultrasonic distance sensor*

This sensor allows us to measure the distance of an object, and in this case is responsible to detect **people crossing**.

- *CO and CO sensor*

These two sensors are responsible to capture, respectively, the **CO** and **CO_2** density in the atmosphere.

- *Luminosity sensor (Photo resistor)*

This sensor allows us to measure the luminosity and to adjust the LED luminosity, according to the light measured in the environment. The data of this sensor is not available in the platform, being its only responsibility to adjust the LED's light intensity accordingly

Coming in accordance with the power source solution we intend to use of the sensor system to acquire and having as example the small sensor system developed in Lisboa-Enova, we made a system sizing for the system developed with the intention of knowing our options regarding this matter. As described before, one of the objectives is for this system to be sustainable, meaning that it would have means to keep operating without the need of being plugged continuously to the electrical grid. For this objective to be achieved, a battery solution have to be considered. To have a better idea of what battery to use, in terms of its capacity, a sizing of the system's current energy expenses is necessary. It is only after these values are known, that the calculation of the necessary capacity of the battery intended can be known. Below we show the results of the energy consumption of the system implemented:

To add a table

After this study, is clear that for the system to be sustainable for at least 24h the battery needed has to have xxxxxxxxxxxx.

Other possibility, and probably the one that will bring the most advantages to the system, is the use of a rechargeable battery powered by a solar panel, making it truly sustainable and stripping it of any dependencies of the electrical grid as it produces its own energy.

A melhorar

However, in the lamp post already converted to LED with CMS would be possible to supply 24h continuous voltage, overcoming these challenges.

9.6 Standards Mapping & Inventory

This annex includes two main elements. The first is an excerpt from the ISO Smart Cities Strategic Advisory Group (SAG) report relating to a specific "deep-dive" example on smart lighting. The second includes a listing of the 353 standards that relate to road lighting or street lighting.

[ISO SAG Report Excerpt] Smart Lighting / Lamppost Standards 'Deep-Dive'

Published standards relating to city lighting have been analysed at two levels:

- Firstly, a run of all ISO-NSB[3] standards in relation to *'road lighting'* or *'street lighting'*. This resulted in 360 standards from 37 standards bodies. Those that were more prominent in production were: AFNOR (FR), BSI (UK), CEN/CENELEC (EU), DIN (DE), DS, NEN (NL), NEMA.
    - Vintage: 1960s – 1; 1970s – 3; 1980s – 12; 1990s – 24; 2000s – 190; 2010s – 130
- Secondly, a more in-depth run of published BSI standards. This was done by searching on origin code, specific classification codes, and key words (lighting system; lumen/illumin, lamp). This resulted in a total of 77 documents. A further filtering on relevance resulted in 5 with high relevance; and 17 with medium relevance. (A favourable comparison to the results from the international scan).

No analysis of "in process" standards has been made.

The initial observations from this rapid and rather coarse exercise suggests:

1. There are a large number of detailed technical standards available; and it would seem few documents that might be considered management frameworks
2. Most standards are national; none are ISO; and few are international
3. There are very few standards that look to system performance (financial / GHG benefits etc)
4. No standards or guidance appear to address integrated infrastructures ('smart' services)

The observation might be that the portfolio of guidance addresses yesterday's market.

City Needs

The market landscape highlights the as yet untapped potential of smart lampposts. Market soundings resulting from the European Innovation Partnership for smart cities (Integrated Infrastructure Action Cluster sub-group: The Humble Lamppost) which is addressing the opportunity of smart lampposts suggests cities are in need of a number of things:

- Understanding of the potential, communicated to the lighting department, and more notably to higher executive levels
- Savings that can result from transition to low energy lighting

[3] *ISO=International Standard Organisation; SDO=Standards Development Organisation; NSB=National Standards Bodies*

- Well-captured and trusted evidence of gains to support the development of their business case

Gap Analysis

The gap however is far greater than is indicated by the majority of cities. There is a rapidly growing number of cities and industry that also see a broader, collective, and altogether bolder set of needs:

- *Willingness of cities to collaborate internally* across functions, to exploit the potential of integrating across infrastructures to use the lamppost as far more than a post for light
- …and to *collaborate externally* between (particularly smaller) cities, to generate scale that will accelerate market uptake and generate economies of scale
- *Common basic solution / service architectures*, and building-block components, to support rapid uptake
- …balanced with *open innovations between service sectors*, to drive innovation
- *Route-maps* that can help build the confidence of cities, to assess where they stand and guide action
- *Investment*, to enable scale replacement of aging lighting assets on a city-wide basis
- *New business models*, to enable greater collaboration between sectors; to reduce the extent by which procurement processes inhibit innovation that can release new forms of value; and to enable new data services to be operationalised

The availability of the above would make a marked improvement in the speed and the scale of market activity, which in turn would deliver substantial benefits in both financial, environmental, and social terms.

Recommendations

In order to support market development in this area a number of recommendations are made:

1. Develop a few leadership guides that make the opportunities that are very apparent clear to city leaders
2. Develop a number of more business-focused 'management frameworks' that can help accelerate decision making and action in the market – addressing the identified gaps above (e.g. solution architecture; business case; business models).
3. Address systems interfaces – physical connectivity, communications, & data – through technical standards
4. In recognition that the existing materials are predominantly nationally produced; and particularly for the upper two tiers of document; consider development of these at an international SDO level; best perhaps as multi-logo documents.
5. Consider rationalising the existing portfolio of standards for this high value topic that offers rapid gains

6. Review ‘in-process’ guidance notably in relation to integrated infrastructures – where the lighting assets may be for multi-purpose use
7. Where possible do the above in a more coordinated manner between SDOs, particularly for management framework documents, to reduce the volume of publications (this type of documents is also less likely to be constrained by national regulations)

Standards Inventory (page 1 of 6)

The publications listed are available from the source organisations (Issuer). Some may attract fees. The list is sorted by issuing (national) standards development organisation (column 3). This listing provides no guarantee of completeness, nor indication of the quality of the documents. Non-EU publications are not listed (e.g. US), however the listing is available.

Document identifier	Publication d	Issuer	Title (English)	Abstract (English)
UNE-EN 40-2:2006	2006-03-01	AENOR (ES)	Lighting columns - Part 2: General requirements and dimensions	
UNE-EN 40-3-1:2013	2013-07-31	AENOR (ES)	Lighting columns - Part 3-1: Design and verification - Specification for characteristic loads	
UNE-EN 40-3-2:2013	2013-07-31	AENOR (ES)	Lighting columns - Part 3-2: Design and verification - Verification by testing	
UNE-EN 40-3-3:2013	2013-07-31	AENOR (ES)	Lighting columns - Part 3-3: Design and verification - Verification by calculation	
UNE-EN 40-4:2006	2006-09-27	AENOR (ES)	Lighting columns - Part 4: Requirements for reinforced and prestressed concrete lighting columns	
UNE-EN 40-5:2003	2003-01-31	AENOR (ES)	Lighting columns. Part 5: Requirements for steel lighting columns.	
UNE-EN 40-6:2003	2003-01-31	AENOR (ES)	Lighting columns. Part 6: Requirements for aluminium lighting columns.	
UNE-EN 40-7:2003	2003-09-05	AENOR (ES)	Lighting columns - Part 7: Requirements for fibre reinforced polymer composite lighting columns	
UNE-EN 13201-2:2004	2004-12-17	AENOR (ES)	Road lighting - Part 2: Performance requirements	
UNE-EN 13201-3:2004	2004-12-17	AENOR (ES)	Road lighting - Part 3: Calculation of performance	
UNE-EN 13201-3:2004/	2007-05-30	AENOR (ES)	Road lighting - Part 3: Calculation of performance	
UNE-EN 13201-4:2005	2005-11-23	AENOR (ES)	Road lighting - Part 4: Methods of measuring lighting performance	
UNE 72401:1981	1981-06-15	AENOR (ES)	LIGHTING COLUMNS. DEFINITION AND TERMS	
UNE 72401/1M:1993	1993-01-22	AENOR (ES)	LIGHTING COLUMNS. DEFINITIONS AND TERMS.	
NF P97-101*NF EN 40-1	1992-03-01	AFNOR	Lighting columns. Part 1 : definitions and terms.	
NF P97-401*NF EN 40-2	2005-05-01	AFNOR	Lighting columns - Part 2 : general requirements and dimensions	
NF P97-405*NF EN 40-3	2014-01-11	AFNOR	Lighting columns - Part 3-1 : design and verification - specification for characteristic loads	
NF P97-406*NF EN 40-3	2014-01-11	AFNOR	Lighting columns - Part 3-3 : design and verification - verification by calculation	
NF P97-407*NF EN 40-3	2014-01-11	AFNOR	Lighting columns - Part 3-2 : design and verification - Verification by testing	
NF P97-412*NF EN 40-4	2006-04-01	AFNOR	Lighting columns - Part 4 : requirements for reinforced and prestressed concrete lighting columns	
NF P97-413*NF EN 40-5	2002-10-01	AFNOR	Lighting columns - Part 5 : requirements for steel lighting columns	
NF P97-414*NF EN 40-6	2002-10-01	AFNOR	Lighting columns - Part 6 : requirements for aluminium lighting columns	
NF P97-415*NF EN 40-7	2004-04-01	AFNOR	Lighting columns - Part 7 : requirements for fibre reinforced polymer composite lighting columns	
FD X90-006-1*FD CEN/	2005-05-01	AFNOR	Road lighting - Part 1 : selection of lighting classes	
NF X90-006-2*NF EN 13	2005-02-01	AFNOR	Road lighting - Part 2 : performance requirements	
PR NF X90-006-2*PR NF		AFNOR	Road lighting - Part 2 : performance requirements	
NF X90-006-3*NF EN 13	2005-02-01	AFNOR	Road lighting - Part 3 : calculation of performance	
PR NF X90-006-3*PR NF		AFNOR	Road lighting - Part 3 : calculation of performance	
NF X90-006-4*NF EN 13	2004-04-01	AFNOR	Road lighting - Part 4 : methods of measuring lighting performance	
PR NF X90-006-4*PR NF		AFNOR	Road lighting - Part 4 : methods of measuring lighting performance	
PR NF X90-006-5*PR NF		AFNOR	Road lighting - Energy performance indicators	
OENORM CEN/TR 1320	2005-09-01	ASI Aus	Road lighting - Part 1: Selection of lighting classes	
OENORM EN 40-1	1992-05-01	ASI Aus	Lighting columns - Definitions and terms	
OENORM EN 40-2	2005-03-01	ASI Aus	Lighting columns - Part 2: General requirements and dimensions	
OENORM EN 40-5	2002-08-01	ASI Aus	Lighting columns - Part 5: Requirements for steel lighting columns	
OENORM EN 40-6	2002-08-01	ASI Aus	Lighting columns - Part 6: Requirements for aluminium lighting columns	
OENORM EN 40-7	2003-05-01	ASI Aus	Lighting columns - Part 7: Requirements for fibre reinforced polymer composite lighting columns	
OENORM EN 13201-2	2013-09-15	ASI Aus	Road lighting - Part 2: Performance requirements	
OENORM EN 13201-2	2004-02-01	ASI Aus	Road lighting - Part 2: Performance requirements	
OENORM EN 13201-3	2013-09-15	ASI Aus	Road lighting - Part 3: Calculation of performance	
OENORM EN 13201-3	2007-06-01	ASI Aus	Road lighting - Part 3: Calculation of performance (consolidated version)	
OENORM EN 13201-4	2013-09-15	ASI Aus	Road lighting - Part 4: Methods of measuring lighting performance	
OENORM EN 13201-4	2004-02-01	ASI Aus	Road lighting - Part 4: Methods of measuring lighting performance	
OENORM EN 13201-5	2013-09-15	ASI Aus	Road lighting - Energy performance indicators	
OENORM O 1051	2007-07-01	ASI Aus	Road lighting - Lighting of conflict areas	
OENORM O 1053	2011-09-15	ASI Aus	Road lighting - Selection of lighting classes - Consideration of the situational traffic flow - National supplements concerning OENORM CEN/TR 13201-1 and OENORM EN 13201-2 to -4	
OEVE/OENORM EN 605	2012-01-01	ASI Aus	Luminaires - Part 2-3: Particular requirements - Luminaires for road and street lighting (IEC 60598-2-3:2002 + A1:2011) (german version)	
BS 5489-1:2013	2012-12-31	BSI	Code of practice for the design of road lighting. Lighting of roads and public amenity areas	
BS 5489-2:2003+A1:20(	2003-12-11	BSI	Code of practice for the design of road lighting. Lighting of tunnels	
BS 5649-5:1982*EN 40-	1982-10-29	BSI	Lighting columns. Specification for base compartments and cableways	Covers post top columns up to 20 m nominal height and columns with brackets up to 18 m nominal height.
BS 5972:1980	1980-09-30	BSI	Specification for photoelectric control units for road lighting	Requirements for photo-electric control units and associated sockets for operation on supply voltages not exceeding 250 V and capable of switching a current not exceeding 10 A. Includes requirements for electrical and photometric performances.
BS 7654:2010	2010-02-28	BSI	Specification for single-phase street lighting cut-out assemblies for low-voltage public electricity distribution systems. 25 A rating for highway power supplies and street furniture	To be read in conjunction with BS EN 60947-1:2007
PD CEN/TR 13201-1:20	2004-08-27	BSI	Road lighting. Selection of lighting classes	
BS EN 40-1:1992	1992-02-28	BSI	Lighting columns. Definitions and terms	Definitions and terms in the field of lighting columns. Includes references to the figures.
BS EN 40-2:2004	2004-10-13	BSI	Lighting columns. General requirements and dimensions	
BS EN 40-3-1:2013	2013-02-28	BSI	Lighting columns. Design and verification. Specification for characteristic loads	
BS EN 40-3-2:2013	2013-02-28	BSI	Lighting columns. Design and verification. Verification by testing	
BS EN 40-3-3:2013	2013-02-28	BSI	Lighting columns. Design and verification. Verification by calculation	
BS EN 40-4:2005	2006-01-16	BSI	Lighting columns. Requirements for reinforced and prestressed concrete lighting columns	
BS EN 40-5:2002	2002-07-17	BSI	Lighting columns. Requirements for steel lighting columns	
BS EN 40-6:2002	2002-07-17	BSI	Lighting columns. Requirements for aluminium lighting columns	
BS EN 40-7:2002	2003-04-29	BSI	Lighting columns. Requirements for fibre reinforced polymer composite lighting columns	
BS EN 13201-2:2003	2003-12-11	BSI	Road lighting. Performance requirements	
BS EN 13201-3:2003	2003-12-11	BSI	Road lighting. Calculation of performance	
BS EN 13201-4:2003	2003-12-11	BSI	Road lighting. Methods of measuring lighting performance	
BS EN 16276:2013	2013-01-31	BSI	Evacuation Lighting in Road Tunnels	
13/30266810 DC	2013-08-19	BSI	BS EN 13201-3. Road lighting. Part 3. Calculation of performance	
13/30271221 DC	2013-08-19	BSI	BS EN 13201-2. Road lighting. Part 2. Performance requirements	
13/30284241 DC	2013-08-19	BSI	BS EN 13201-4. Road lighting. Part 4. Methods of measuring lighting performance	
13/30284244 DC	2013-08-19	BSI	BS EN 13201-5. Road lighting. Energy performance indicators	
EN 40-2	2004-10-00	CEN	Lighting columns - Part 2: General requirements and dimensions	

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EN 40-3-1	2013-02-00	CEN	Lighting columns - Part 3-1: Design and verification - Specification for characteristic loads	This European Standard specifies design loads for lighting columns. It applies to lighting columns of nominal height (including any bracket) not exceeding 20 m. Special structural designs to permit the attachment of signs, overhead wires, etc. are not covered by this European Standard. The requirements for lighting columns made from materials other than concrete, steel, aluminium or fibre reinforced polymer composite (for example wood, plastic and cast iron) are not specifically covered in this standard. Fibre reinforced polymer composite lighting columns are covered in this document, in conjunction with Annex B of EN 40-7:2002. This European Standard includes performance requirements for horizontal loads due to wind. Passive safety and the behaviour of a lighting column under the impact of a vehicle are not addressed. Such lighting columns will have additional requirements (see EN 12767).
EN 40-3-2	2013-02-00	CEN	Lighting columns - Part 3-2: Design and verification - Verification by testing	This European Standard specifies the requirements for the verification of the design of steel, aluminium, concrete and fibre reinforced polymer composite lighting columns by testing. It gives type tests and so does not cover testing for quality control purposes. It applies to lighting columns of nominal height (including any bracket) not exceeding 20 m. Special structural designs to permit the attachment of signs, overhead wires, etc. are not covered by this European Standard. This European Standard includes a simplified method for testing steel and aluminium lighting columns. Refer to EN 40-4 for concrete lighting columns and to EN 40-7 for fibre reinforced polymer composite lighting columns. NOTE For a more detailed test procedure, refer to Annex D of EN 1990:2002. The requirements for lighting columns made from materials other than concrete, steel, aluminium or fibre reinforced polymer composite (for example wood, plastic and cast iron) are not specifically covered in this European Standard. This European Standard includes performance requirements for horizontal loads due to wind. Passive safety and the behaviour of a lighting column under the impact of a vehicle are not addressed. Such lighting columns will have additional requirements (see EN 12767).
EN 40-4	2005-12-00	CEN	Lighting columns - Part 4: Requirements for reinforced and prestressed concrete lighting columns	
EN 40-4/AC	2006-09-00	CEN	Lighting columns - Part 4: Requirements for reinforced and prestressed concrete lighting columns	
EN 40-5	2002-04-00	CEN	Lighting columns - Part 5: Requirements for steel lighting columns	
EN 40-6	2002-04-00	CEN	Lighting columns - Part 6: Requirements for aluminium lighting columns	
EN 40-7	2002-12-00	CEN	Lighting columns - Part 7: Requirements for fibre reinforced polymer composite lighting columns	
FprCEN/TR 13201-1	2013-08-00	CEN	Road lighting - Part 1: Guidelines on selection of lighting classes	The decision on whether a road should be lit is defined in the national road lighting policy. This varies by country or municipality. Specific guidelines are usually available at national level for each country. This Technical Report does not give the criteria on which a decision to light an area can be made, nor on how a lighting installation should be used. Further guidance is given in CIE 115:2010 (Paragraph and Annex A). This Technical Report specifies the lighting classes set out in EN 13201-2 and gives guidelines on the selection of the most appropriate class for a given situation. To do this, it includes a system to define appropriate lighting classes for different outdoor public areas in terms of parameters relevant to guarantee the aims presented in introductions. The methods presented in Clauses 5, 6 and 7 have to be considered as the starting points of a comprehensive approach for the normal road lighting. In that sense, the models cannot cover all the different road cases; they introduce general parameters and the impact on lighting requirements. Only the real situation and its unique characteristics (geometry of the road, marking, visual environment, difficulty of the navigation task, lack of visibility, risks of glares due to existing elements, local weather, specific users such as high rate of elderly or visually impaired people, etc.) can lead to a final determination of the appropriate lighting class applying risk evaluation techniques. The visual needs of road users under reduced traffic volumes during certain periods of night or under varying weather conditions, and the positive benefits of reduced energy consumption and potential environmental improvements, are some of the considerations which justify the installation of adaptive road lighting. There are a variety of suitable instruments, devices and methods which can be used for the intelligent control of a road lighting installation. The control systems range from very simple to the most sophisticated applications. Annex B is of assistance in choosing the correct lighting level when adaptive lighting is used as it provides a more refined evaluation of the luminance or illuminance levels within the specific lighting class. Whilst the luminance or illuminance levels may be varied to suit reduced traffic volumes, weather conditions or other parameters the quality parameters of the applicable lighting class specified in EN 13201-2 should be maintained at all times. Renewal or refurbishment of obsolete and uneconomic installations is important. It may be possible to obtain more adapted lighting levels with lower energy consumption using new designs and new technology. The upgrading of lighting and control systems will often give good cost-benefit ratios and short amortisation periods. This document does not give guidelines on the selection of lighting classes for toll stations, tunnels or canals and locks.
CEN/TR 13201-1	2004-07-00	CEN	Road lighting - Part 1: Selection of lighting classes	
prEN 13201-2	2013-08-00	CEN	Road lighting - Part 2: Performance requirements	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting. NOTE Installed intensity classes for the restriction of disability glare and control of obtrusive light and installed glare index classes for the restriction of discomfort glare are defined in the informative annex A.
EN 13201-2	2003-11-00	CEN	Road lighting - Part 2: Performance requirements	
prEN 13201-3	2013-08-00	CEN	Road lighting - Part 3: Calculation of performance	This European Standard defines and describes the conventions and mathematical procedures to be adopted in calculating the photometric performance of road lighting installations designed in accordance with the parameters described in EN 13201-2 to ensure that every lighting calculation is based on the same mathematical principles. The design procedure of a lighting installation requires also the knowledge of the parameters involved in the described model, their tolerances and variability. These aspects are not considered in this part of the standard but a procedure to analyse their contribution in the expected results is suggested in part 4 and it can be used in the design phase too.
EN 13201-3	2003-11-00	CEN	Road lighting - Part 3: Calculation of performance	
EN 13201-3/AC	2007-02-00	CEN	Road lighting - Part 3: Calculation of performance	
prEN 13201-4	2013-08-00	CEN	Road lighting - Part 4: Methods of measuring lighting performance	This European Standard specifies measurement conditions and procedures for measuring the quality parameters of road lighting installations i.e. the parameters that quantifies their performances according to EN 13201-2.
EN 13201-4	2003-11-00	CEN	Road lighting - Part 4: Methods of measuring lighting performance	
prEN 13201-5	2013-08-00	CEN	Road lighting - Energy performance indicators	This part of this European Standard defines how to calculate the energy performance indicators for road lighting installations using the calculated power density (D) and the calculated energy consumption indicator (ECIy). Power density (D) demonstrates the energy needed for a road lighting installation, while it is fulfilling the relevant lighting requirements specified in EN 13201-2. The energy consumption indicator (ECIy) determines the power consumption during the year, even if the relevant lighting requirements change during the night or seasons. These indicators may be used to compare the energy performance of different road lighting solutions and technologies for the same road lighting project. The energy performance of road lighting systems with different road geometries or different lighting requirements cannot be compared to each other directly, as the energy performance is influenced by, amongst others, the geometry of the area to be lit, as well as, the lighting requirements. The power density (D) and energy consumption indicators (ECIy) apply for all traffic areas covered by the series of lighting classes M, C and P as defined in EN 13201-2. Annex B introduces the installation efficacy and its factors as a measure of the influence of various losses and parameters.
CR 14380	2003-04-00	CEN	Lighting applications - Tunnel lighting	
EN 60598-2-3	2003-02-00	CENELEC	Luminaires - Part 2-3: Particular requirements; Luminaires for road and street lighting (IEC 60598-2-3:2002)	
EN 60598-2-3/AC	2005-08-00	CENELEC	Luminaires - Part 2-3: Particular requirements; Luminaires for road and street lighting (IEC 60598-2-3:2002)	
EN 60598-2-3/A1	2011-05-00	CENELEC	Luminaires - Part 2-3: Particular requirements - Luminaires for road and street lighting (IEC 60598-2-3:2002/A1:2011)	
CIE 32*CIE 32.1	1977-00-00	CIE Internatr	Lighting in situations requiring special treatment	This technical report supplements CIE 12.2-1977 Recommendations for the Lighting of Roads for Motorized Traffic. It describes details of technologies and work processes which are required to achieve the objectives put forward in CIE 12.2-1977. It treats lighting problems which arise form certain discontinuities not mentioned in CIE 12.2-1977 (e.g. humps or multi-level works). It also takes into account the evolution of newer techniques (e.g. new sources, high masts, or catenary lighting).
CIE 33*CIE 33.1	1977-00-00	CIE Internatr	Depreciation of Installation and Their Maintenance (in Road Lighting)	This report is supplementary to CIE 12.2-1977 Recommendations for the Lighting of Roads for Motorized Traffic. The first part contains a survey of the combined effects of permanent or incidental causes which progressively impair the performance of road lighting installations. The second part points out what preventive measures can be taken in designing and installing a system to reduce or slow down the deterioration of materials (columns, lanterns, light sources, auxiliary gear). The last part indicates preventive maintenance operations aimed at protecting materials against deterioration or at restoring, as much as possible, the initial performance of the installation. The different methods of maintenance and replacement of various items are analyzed.
CIE 61	1984-00-00	CIE Internatr	Tunnel entrance lighting; a survey of fundamentals for determining the luminance in the threshold zone	The purpose of this report is to provide information on the lighting requirements for drivers approaching the entrance of a tunnel in daytime. The report is restricted to the first stretch of tunnels, called the threshold zone. The report begins with a review of the fundamental experiments that were the basis of the 1973 CIE recommendations for tunnel lighting, as far as the threshold zone is concerned (see CIE 23-1973 International Recommendations for Motorway Lighting). New investigations are described with which the original fundamental experiments may be applied more realistically. Results of driving experiments in a real tunnel under various lighting conditions are summarized. To complete the description of methods based on visibility, two recently published methods that dispense with the original fundamental experiments are discussed. Finally, mention is made of an approach not directly based on visibility but on traffic flow. The report does not give a straightforward answer to all problems related to the lighting of threshold zones of tunnels. It summarizes the state of the art and, as such, will form the background for future tunnel lighting recommendations.

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CIE 93	1992-00-00	CIE Internatr	Road lighting as an accident countermeasure	Road accidents at night are disproportionately high in numbers and severity compared to day. The major factor contributing to this problem is darkness because of its great influence on the driver's behaviour and ability. At night man's visual capabilities are impaired and visibility is reduced. Thus road lighting is a potential countermeasure. Sixty-two lighting and accident studies, from 15 countries, have been rigourously analysed. Some 85% of results show lighting to be beneficial, with about one third of these having statistical significance. These lead to the general conclusion that road lighting on traffic routes will reduce the incidence of night accidents. Depending on the class of road and the accident classification involved, the statistically significant results show reductions of between 13% and 75%. Depending on the road and traffic classification, cost benefit studies show the accident savings to more than offset the cost of lighting. In the report, warrants for lighting related to accidents and traffic volume are developed. The findings can assist in the preparation of national standards and in decisions onthe implementation of road lighting programmes.
CIE 132	1999-00-00	CIE Internatr	Design methods for lighting of roads	Calculation methods for designing road lighting installations, which satisfy certain quality criteria, are provided.
CIE 136	2000-00-00	CIE Internatr	Guide to the lighting of urban areas	The purpose of this guide is to supplement the lightning recommendations and standards for roads and areas of public.
CIE 140	2000-00-00	CIE Internatr	Road lighting calculations	It includes th calculation of luminance, illuminance and their associated measures of uniformity, as well as disability glare.
CIE 180	2007-00-00	CIE Internatr	Road transport lighting for developing countries	This Technical Report has been prepared by CIE Technical Committee 4-37 of Division 4 "Lighting and Signalling for Transport" and has been approved by the Board of Administration of the Commission Internationale de l'Eclairage for study and application. The document reports on current knowledge and experience within the specific field of light and lighting described, and is intended to be used by the CIE membership and other interested parties. It should be noted, however, that the status of this document is advisory and not mandatory. The latest CIE proceedings or CIE NEWS should be consulted regarding possible subsequent amendments.
CIE 189	2010-00-00	CIE Internatr	Calculation of tunnel lighting quality criteria	Experience of making road tunnel lighting designs in accordance with publication CIE 88:2004 has shown that there are some aspects of calculation of the different lighting criteria where more specific guidance to the designers is necessary. This document explains how the tunnel environment differs significantly from the open road situation. In particular, the presence of walls along the traffic road involves reflection effects between different surfaces. The variation of luminance level along the tunnel, the changes in the lighting installation along the entrance section of the tunnel and the use of different lighting systems in different parts of the tunnel introduce more complexity in the lighting calculations. The report gives guidance on determining the method for calculating the relevant lighting quality criteria for tunnel situations.
CIE 194	2011-00-00	CIE Internatr	On site measurement of the photometric properties of road and tunnel lighting	This Technical Report gives the information which designers and users of automatic systems intended to measure the photometric properties of road and tunnel lighting systems need in order to understand the performance of such systems. The main aim of this guide is to clearly define and describe the technical limitations and characteristics of these measurement systems. In this way the advantages and disadvantages of the technical solutions adopted in different systems can be assessed. Their measurement uncertainty can be evaluated and discrepancies understood when the measurement results are compared with those obtained by manual systems or by theoretical evaluations from mathematical algorithms and/or computer simulations. The guide concentrates on systems that measure horizontal illuminance and luminance on road surfaces but much of the guidance is relevant to other quantities and measurement conditions. The guide is divided into two parts: - Review of essential photometric quantities where differences between their physical or normative definition, their mathematical evaluation and their measurable approximation are highlighted (Clause 2 and Clause 3); - Review of the types and characteristics of instruments, the conditions and methodologies of measurements, the evaluation of uncertainty in measurement results and data elaboration (Clause 5, Clause 6 and Clause 7).
CSN CEN/TR 13201-1	2007-03-01	CNI Cz	Road lighting - Part 1: Selection of lighting classes	
CSN EN 13201-4	2005-05-01	CNI Cz	Road lighting - Part 4: Methods of measuring lighting performance	
CSN EN 13201-3	2005-05-01	CNI Cz	Road lighting - Part 3: Calculation of performance	
CSN EN 13201-2	2005-05-01	CNI Cz	Road lighting - Part 2: Performance requirements	
CSN EN 60598-2-3 ed. 2	2003-09-01	CNI Cz	Luminaires - Part 2-3: Particular requirements - Luminaires for road and street lighting	
CAN/CSA-E60598-2-3-9	1998-06-01	CSA Can	Luminaires - Part 2: Particular Requirements - Section 3: Luminaires for Road and Street Lighting	This National Standard of Canada is based on International Standard CEI/IEC 598-2-3: 1993 (second edition, 1993-04), including Amendment 1:1997, and 2:2000. Note - Amendment No. 2 reaffirmed 2007-03-28. Scope This Standard specifies requirements for luminaires for road and street lighting for use with tungsten filament, tubular fluorescent, and other discharge lamps designed to be installed on branch circuits not exceeding 600 V nominal or less between conductors in accordance with the rules of CSA Standard C22.1-02, Canadian Electrical Code, Part I. It is to be read in conjunction with those clauses of CSA Standard CAN/CSA-E60598-1 to which reference is made.
LiTG Publikation 7	1977-00-00	De Light	Measures of road lighting effectiveness - Transactions	
DIN 5042-4	1994-03-00	DIN	Combustion lamps and gas luminaires; incandescent mantle for roadlighting; requirements and testing	The document states the requirements for incandescent mantles used in gas luminaires for street lighting regarding luminous flux output, shockproofness, fitness for transport. Additionally the relevant test methods are given.
DIN 13201-1	2005-11-00	DIN	Road lighting - Part 1: Selection of lighting classes; German version CEN/TR 13201-1:2004	This document specifies the lighting classes set out in EN 13201-2 and gives guidelines on the application of these classes. It also gives guidelines on the selection of the relevant area to which the lighting classes from EN 13201-2 and the calculation grids and procedure from EN 13201-3 should be applied. This document does not give the criteria on which a decision to light an area can be made, nor on how a lighting installation should be used. This document does not give guidelines on the selection of lighting classes for toll stations, tunnels or canals and locks.
DIN 67523-1	2010-06-00	DIN	Lighting of pedestrian crossings (sign 293 StVO) with additional lighting - Part 1: General quality characteristics and guide values	This standard specifies the minimum lighting requirements which, in case of their nonfulfilment, make an additional lighting of pedestrian crossings necessary. At the same time it specifies, how such an additional lighting shall be designed in order to secure that the pedestrian who has the right of way is seen clearly and in time on the pedestrian crossing by the other traffic participant.
DIN 67523-2	2010-06-00	DIN	Lighting of pedestrian crossings (sign 293 StVO) with additional lighting - Part 2: Calculation and measurement	This standard specifies the measurement and calculation of the quantitative photometric quality criteria according to DIN 67523-1. The specifications are necessary for comparable calculations and consistent measurements.
DIN 67524-1	2008-07-00	DIN	Lighting of street tunnels and underpasses - Part 1: General quality characteristics and guide values	The document applies to lighting of tunnels and underpasses with motor traffic. The lighting of tunnels is important for an easy traffic flow and the safety.
DIN 67524-2	2011-06-00	DIN	Lighting of road tunnels and underpasses - Part 2: Calculation and measurement	This standard specifies the calculation and measurement procedure of illumination quality criteria of road tunnels according to DIN 67524-1. This standard includes among others specifications on planning and calculation of road tunnel lighting as well as measurement and evaluation of essential lighting quantities. Further specifications of the control of the illumination in the entrance, transition and interior zone of road tunnels are given.
DIN EN 40-2	2005-02-00	DIN	Lighting columns - Part 2: General requirements and dimensions; German version EN 40-2:2004	This European Standard specifies the requirements and dimensions for lighting columns, brackets, base compartments, cableways and earthing terminals. It applies to post top columns not exceeding 20 m height for post top lanterns and columns with brackets not exceeding 18 m height for side entry lanterns. This Part does not attempt to restrict the actual appearance or shape of the column or bracket.
DIN EN 40-3-1	2013-06-00	DIN	Lighting columns - Part 3-1: Design and verification - Specification for characteristic loads; German version EN 40-3-1:2013	This European Standard specifies design loads for lighting columns. It applies to lighting columns of nominal height (including any bracket) not exceeding 20m. Special structural designs to permit the attachment of signs, overhead wires, etc. are not covered by this standard. The requirements for lighting columns made from materials other than concrete, steel, aluminium or fibre reinforced polymer composite (for example wood, plastic and cast iron) are not specifically covered in this standard. Fibre reinforced polymer composite lighting columns are covered in this document, in conjunction with Annex B of EN 40-7.This standard includes performance requirements for horizontal loads due to wind. Passive safety and the behavior of a lighting column under the impact of a vehicle are not addressed. Such lighting columns will have additional requirements (see EN 12767).
DIN EN 40-3-2	2013-06-00	DIN	Lighting columns - Part 3-2: Design and verification - Verification by testing; German version EN 40-3-2:2013	This European standard specifies the requirements for the verification of the design of steel, aluminium, concrete and fibre reinforced polymer composite lighting columns by testing. It gives type tests and so does not cover testing for quality control purposes. It applies to lighting columns of nominal height (including any bracket) not exceeding 20m. Special structural designs to permit the attachment of signs, overhead wires, etc. are not covered by this standard.NOTE: This Standard includes only a simplified method. For a more detailed test procedure refer to Annex D of EN 1990.The requirements for lighting columns made from materials other than concrete, steel, aluminium or fibre reinforced polymer composite (for example wood, plastic and cast iron) are not specifically covered in this standard. Fibre reinforced polymer composite lighting columns are covered in conjunction with Annex B of EN 40-7.This standard includes performance requirements for horizontal loads due to wind. Passive safety and the behavior of a lighting column under the impact of a vehicle are not addressed. Such lighting columns will have additional requirements (see EN 12767).
DIN EN 40-4	2006-06-00	DIN	Lighting columns - Part 4: Requirements for reinforced and prestressed concrete lighting columns; German version EN 40-4:2005	
DIN EN 40-4 Berichtigu	2008-05-00	DIN	Lighting columns - Part 4: Requirements for reinforced and prestressed concrete lighting columns; German version EN 40-4:2005; Corrigenda to DIN EN 40-4:2006-06, German version EN 40-4:2005/AC:2006	This document specifies requirements for reinforced and prestressed concrete lighting columns. It applies to columns not exceeding 20 m height for post top lanterns and columns with brackets not exceeding 18 m height for side entry lanterns.
DIN EN 40-5	2002-09-00	DIN	Lighting columns - Part 5: Requirements for steel lighting columns; German version EN 40-5:2002	This European Standard specifies requirements for steel lighting columns. It includes materials and conformity control. It applies to post top columns not exceeding 20 m height for post top lanterns and to columns with brackets not exceeding 18 m height for side entry lanterns.
DIN EN 40-6	2002-09-00	DIN	Lighting columns - Part 6: Requirements for aluminium lighting columns; German version EN 40-6:2002	This European Standard specifies requirements for aluminium lighting columns. It includes materials and conformity control. It applies to post top columns not exceeding 20 m height for post top lanterns and to columns with brackets not exceeding 18 m height for side entry lanterns.
DIN EN 40-7	2003-06-00	DIN	Lighting columns - Part 7: Requirements for fibre reinforced polymer composite lighting columns; German version EN 40-7:2002	The document specifies the performance requirements for fibre reinforced polymer composite lighting columns for which the main intended use is road lighting. It includes materials and test methods. The composite materials considered are those constructed from a fibrous reinforcing material that is suspended in a matrix of resin material. It applies to post top columns not exceeding 20 m height for post top lanterns and columns with brackets not exceeding 18 m height for side entry lanterns.

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DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
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DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DIN EN 13201-2	2013-09-00	DIN	Road lighting - Part 2: Performance requirements; German version prEN 13201-2:2013	This part of this European Standard defines lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
DS/EN 60598-2-3/Corr.	2006-04-26	DS	Luminaires - Part 2-3: Particular requirements - Luminaires for road and street lighting	This part of 60598 specifies requirements for - luminaires for road, street lighting and other public outdoor lighting applications; - tunnel lighting; - column-integrated luminaires with a minimum total height above normal ground level of 2.5 m; and for use with electrical lighting sources on supply voltages not exceeding 1000V. NOTE Column integrated luminaires with a total height below 2,5 m are under consideration.
DS/EN 60598-2-3	2005-09-29	DS	Luminaires - Part 2-3: Particular requirements - Luminaires for road and street lighting	This part of 60598 specifies requirements for - luminaires for road, street lighting and other public outdoor lighting applications; - tunnel lighting; - column-integrated luminaires with a minimum total height above normal ground level of 2.5 m; and for use with electrical lighting sources on supply voltages not exceeding 1000V. NOTE Column integrated luminaires with a total height below 2,5 m are under consideration.
LST EN 13201-4:2004/P	2013-07-15	LST (Lt)	Road lighting - Part 4: Methods of measuring lighting performance	
LST EN 13201-3:2004/A	2013-07-15	LST (Lt)	Road lighting - Part 3: Calculation of performance	
LST EN 13201-3:2004/P	2013-07-15	LST (Lt)	Road lighting - Part 3: Calculation of performance	
LST EN 13201-3:2004/A	2007-07-13	LST (Lt)	Road lighting - Part 3: Calculation of performance	
LST EN 13201-3:2004	2004-06-14	LST (Lt)	Road lighting - Part 3: Calculation of performance	
LST EN 13201-4:2004	2004-06-14	LST (Lt)	Road lighting - Part 4: Methods of measuring lighting performance	
LST EN 13201-2:2004	2004-06-14	LST (Lt)	Road lighting - Part 2: Performance requirements	
LST CEN/TR 13201-1:20	2010-02-26	LST (Lt)	Road lighting - Part 1: Selection of lighting classes	
LST EN 60598-2-3:2003	2003-12-28	LST (Lt)	Luminaires. Part 2-3: Particular requirements. Luminaires for road and street lighting (IEC 60598-2-3:2002)	
NBN L 18-001:1980	1980-01-00	NBN	Code of practice for public road lighting	
NBN L 18-002:1988	1988-01-00	NBN	Recommendations for special cases of public lighting	
NBN L 18-004:2012	2012-02-00	NBN	Road lighting - Selection of lighting classes	
NBN S 28-002:1982	1982-01-00	NBN	Lighting columns - Dimensions and tolerances	
NBN S 28-003:1986	1986-01-00	NBN	Lighting columns - Materials	
NBN S 28-004:1986	1986-01-00	NBN	Lighting columns - Surface protection of metal lighting columns	
NBN S 28-009:1986	1986-01-00	NBN	Lighting columns - Special requirements for reinforced and prestressed concrete lighting columns	
NBN EN 40-2:2004	2004-12-20	NBN	Lighting columns - Part 2: General requirements and dimensions	
NBN EN 40-4:2006	2006-05-05	NBN	Lighting columns - Part 4 : Requirements for reinforced and prestressed concrete lighting columns (+ AC:2006)	
NBN EN 40-5:2000	2000-07-00	NBN	Lighting columns - Part 5: Specification for steel lighting columns	
NBN EN 40-5:2002	2002-09-11	NBN	Lighting columns - Part 5: Requirements for steel lighting columns	
NBN EN 40-6:2000	2000-07-00	NBN	Lighting columns - Part 6: Specification for aluminium lighting columns	
NBN EN 40-6:2002	2002-09-11	NBN	Lighting columns - Part 6: Requirements for aluminium lighting columns	
NBN EN 40-7:2003	2003-03-28	NBN	Lighting columns - Part 7: Requirements for fibre reinforced polymer composite lighting columns	
CEN/TR 13201-1		NBN	Road lighting - Part 1: Selection of lighting classes	
NBN EN 13201-2:2004	2004-01-14	NBN	Road lighting - Part 2: Performance requirements	
NBN EN 13201-3:2004	2004-01-14	NBN	Road lighting - Part 3: Calculation of performance (+AC:2005+2007)	
NBN EN 13201-4:2004	2004-01-14	NBN	Road lighting - Part 4: Methods of measuring lighting performance	
NEN-EN 40-3-3:2013 en	2013-04-01	NEN	Lighting columns - Part 3-3: Design and verification - Verification by calculation	This European Standard specifies the requirements for the verification of the design of lighting columns by calculation. It applies to lighting columns of nominal height (including any bracket) not exceeding 20 m. Special structural designs to permit the attachment of signs, overhead wires, etc. are not covered by this European Standard. The requirements for lighting columns made from materials other than concrete, steel, aluminium or fibre reinforced polymer composite (for example wood, plastic and cast iron) are not specifically covered in this standard. Fibre reinforced polymer composite lighting columns are covered in this standard in conjunction with EN 40-7. This European Standard includes performance requirements for horizontal loads due to wind. Passive safety and the behaviour of a lighting column under the impact of a vehicle are not addressed. Such lighting columns will have additional requirements (see EN 12767). The calculations used in this European Standard are based on limit state principles, where the effects of factored loads are compared with the relevant resistance of the structure. Two limit states are considered: a) the ultimate limit state, which corresponds to the load-carrying capacity of the lighting column; b) the serviceability limit state, which relates to the deflection of the lighting column in service.

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NEN-EN-IEC 60598-2-3	2011-06-01	NEN	Luminaires - Part 2-3: Particular requirements - Luminaires for road and street lighting	
NEN-EN 40-3-2:2013 er	2013-04-01	NEN	Lighting columns - Part 3-2: Design and verification - Verification by testing	This European standard specifies the requirements for the verification of the design of steel, aluminium, concrete and fibre reinforced polymer composite lighting columns by testing. It gives type tests and so does not cover testing for quality control purposes. It applies to lighting columns of nominal height (including any bracket) not exceeding 20 m. Special structural designs to permit the attachment of signs, overhead wires, etc. are not covered by this European Standard. This European Standard includes a simplified method for testing steel and aluminium lighting columns. Refer to EN 40-4 for concrete lighting columns and to EN 40-7 for fibre reinforced polymer composite lighting columns. The requirements for lighting columns made from materials other than concrete, steel, aluminium or fibre reinforced polymer composite (for example wood, plastic and cast iron) are not specifically covered in this European Standard. This European Standard includes performance requirements for horizontal loads due to wind. Passive safety and the behaviour of a lighting column under the impact of a vehicle are not addressed. Such lighting columns will have additional requirements.
NEN-EN-IEC 60598-2-3	2003-04-01	NEN	Luminaires - Part 2-3: Particular requirements - Luminaires for road and street lighting	Specifies requirements for luminaires for road and street lighting, dor use with tungsten filament, tubular fluorescent and other discharge lamps on supply voltages not exceeding 1000 V.
NEN-EN 40-3-1:2013 er	2013-04-01	NEN	Lighting columns - Part 3-1: Design and verification - Specification for characteristic loads	This European Standard specifies design loads for lighting columns. It applies to lighting columns of nominal height (including any bracket) not exceeding 20 m. Special structural designs to permit the attachment of signs, overhead wires, etc. are not covered by this European Standard. The requirements for lighting columns made from materials other than concrete, steel, aluminium or fibre reinforced polymer composite (for example wood, plastic and cast iron) are not specifically covered in this standard. Fibre reinforced polymer composite lighting columns are covered in this document, in conjunction with Annex B of EN 40-7:2002. This European Standard includes performance requirements for horizontal loads due to wind. Passive safety and the behaviour of a lighting column under the impact of a vehicle are not addressed. Such lighting columns will have additional requirements (see EN 12767).
NEN-EN 16276:2013 en	2013-01-01	NEN	Evacuation Lighting in Road Tunnels	This European Standard specifies evacuation lighting in road tunnels longer than 500 m and with an AADT (Annual Average Daily Traffic) higher than 500 vehicles to facilitate the safe evacuation of vehicle occupants in evacuation situations such as fire. It addresses the fundamental issues of evacuation lighting for evacuation routes, emergency exits and cross connections, as well as giving some practical advice regarding aspects of installation and maintenance in road tunnels. It is intended to be used in conjunction with CEN/CR 14380:2003 or relevant national standards for road tunnel lighting. The recommendations may be applied to tunnels up to 500 m in length, especially where conditions such as high traffic volume, or severe curvature or gradient apply. It specifies lighting levels and general provisions for evacuation lighting installations that, based on experience, are considered to be necessary for the safety of people driving through road tunnels in case of an incident and particularly in case of fire. However, as there are different types of road tunnels, both in construction and traffic conditions, various types of incident may occur. This standard should be considered as a list of minimum prescriptions for evacuation lighting in tunnels, to be completed by means of specific risk analysis for the particular tunnel. The design of marking and safety signs is not part of this standard.
NEN-EN 13201-3:2003/	2007-03-01	NEN	Road lighting - Part 3: Calculation of performance	
NEN-EN 40-4:2005/C1:	2007-02-01	NEN	Lighting columns - Part 4: Requirements for reinforced and prestressed concrete lighting columns	
NEN-EN 13201-3:2003/	2006-09-01	NEN	Road lighting - Part 3: Calculation of performance	
NEN-EN 40-4:2005 en	2005-12-01	NEN	Lighting columns - Part 4: Requirements for reinforced and prestressed concrete lighting columns	This document specifies requirements for reinforced and prestressed concrete lighting columns. It applies to columns not exceeding 20 m height for post top lanterns and columns with brackets not exceeding 18 m height for side entry lanterns. This document specifies: a) performance related to the essential requirement of resistance to horizontal (wind) loads, measured in accordance with EN 40-3; b) performance under vehicle impact (passive safety) in support of the Essential Requirement No. 4 Safety in use, measured in accordance with the corresponding test methods included in this document or available in separate European Standards.
NEN-EN-IEC 60598-2-3	2005-09-01	NEN	Luminaires - Part 2-3: Particular requirements - Luminaires for road and street lighting	
NPR-CEN/TR 13201-1:2	2004-08-01	NEN	Road lighting - Part 1: Selection of lighting classes	This document specifies the lighting classes set out in EN 13201-2 and gives guidelines on the application of these classes. To do this, it includes a system to define an outdoor public traffic area in terms of parameters relevant to lighting. To assist in the application of classes, it suggests a practical relationship between the various series of lighting classes, in terms of comparable or alternative classes. It also gives guidelines on the selection of the relevant area to which the lighting classes from EN 13201-2 and the calculation grids and procedure from EN 13201-3 should be applied. The parameters used in this document allow: a) a lighting situation to be described in terms of: - the geometry of the area under consideration; - the use of the area; - the influence of the surrounding environment; b) a specific approach to situations to be taken to enable the effective use of energy. This document does not give the criteria on which a decision to light an area can be made, nor on how a lighting installation should be used. This document does not give guidelines on the selection of lighting classes for toll stations, tunnels or canals and locks.
NEN-EN 13201-2:2003	2003-12-01	NEN	Road lighting - Part 2: Performance requirements	This part of this European Standard defines, according to photometric requirements, lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting.
NEN-EN 13201-4:2003	2003-12-01	NEN	Road lighting - Part 4: Methods of measuring the lighting performance	This part of this European Standard specifies the procedures for making photometric and related measurements of road lighting installations. Examples are given of the form of the test report.
NEN-EN 13201-3:2003	2003-12-01	NEN	Road lighting - Part 3: Calculation of performance	This European Standard defines and describes the conventions and mathematical procedures to be adopted in calculating the photometric performance of road lighting installations designed in accordance with EN 13201-2.
NEN-EN 40-7:2003 en	2003-02-01	NEN	Lighting columns - Part 7: Requirements for fibre reinforced polymer composite lighting columns	This part of this European Standard specifies the performance requirements for fibre reinforced polymer composite lighting columns for which the main intended use is road lighting. It includes materials and test methods. The composite materials considered are those constructed of a resin matrix reinforced by a high strength fibrous material. It applies to post top columns not exceeding 20 m height for post top lanterns and columns with brackets not exceeding 18 m height for side entry lanterns. This European Standard specifies the classes of performance related to the essential requirements of resistance to horizontal (wind) loads, and performance under vehicle impact (passive safety) in application of Essential Requirement No 4 Safety in Use measured according to the corresponding test methods included in this European Standard or available in separate European Standards. It provides for the evaluation of conformity of the product to this European Standard.
NPR 13201-1:2002 nl	2002-05-01	NEN	Public lighting - Part 1: Quality criteria	This NPR gives guidelines for the choice of road lighting that matches as good as possible to the outdoor public traffic area situation. It defines lighting classes for road lighting aiming at the visual needs of road users. They are related to the technical aspects of road utilization and behaviour in different traffic situations.
NEN-EN 40-5:2002 en	2002-05-01	NEN	Lighting columns - Part 5: Requirements for steel lighting columns	Specifies requirements for steel lighting columns. It includes materials and conformity control. It applies to post top columns not exceeding 20 m height for post top lanterns and to columns with brackets not exceeding 18 m height for side entry lanterns. Specifies performance related to the essential requirements of resistance to horizontal (wind) loads and performance under vehicle impact (passive safety) in support of the Essential Requirement No 4 Safety in use measured according to the corresponding test methods included in this European Standard or available in separate European Standards. It provides for the evaluation of conformity of the products to this European Standard.
NEN-EN 40-6:2002 en	2002-05-01	NEN	Lighting columns - Part 6: Requirements for aluminium lighting columns	Specifies requirements for aluminium lighting columns. It includes materials and conformity control. It applies to post top columns not exceeding 20 m height for post top lanterns and columns with brackets not exceeding 18 m height for side entry lanterns. Specifies performance related to the essential requirements of resistance to horixontal (wind) loads and performance under vehicle impact (passive safety) in support of the Essential Requirement No 4 Safety in use measured according to the corresponding test methods included in this European Standard or available in separate European Standards. It provides for the evaluation of conformity of the products to this European Standard.
NEN-EN 40-2:2004 en	2004-10-01	NEN	Lighting columns - Part 2: General requirements and dimensions	This document specifies the requirements and dimensions for lighting columns, brackets, base compartments, cableways and earthing terminals. It applies to post top columns not exceeding 20 m height for post top lanterns and columns with brackets not exceeding 18 m height for side entry lanterns. This Part does not attempt to restrict the actual appearance or shape of the column or bracket. The majority of lighting columns are normally of a stepped tubular, round, octagonal or polygonal crosssection. Lighting columns may be manufactured from materials other than those listed in the foreword (e.g. wood, plastic, cast iron) or in other forms (e.g. lattice and telescopic). This document specifies performance related to the essential requirements of resistance to horizontal (wind) loads and performance under vehicle impact (passive safety) in support of the Essential Requirement No 4 Safety in use measured according to the corresponding test methods included in this document or available in separate documents.

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NPR-CR 14380:2005 en	2005-07-01	NEN	Lighting applications - Tunnel lighting	This CEN Technical Report gives guidance on the design of the lighting of road tunnels and underpasses for motorized and mixed traffic. This guidance concerns arrangements, levels and other parameters including daylight, which are related only to traffic safety. Aspects concerning visual comfort should be chosen in agreement with national practice. The guidance in this report may be applied to any tunnel or underpass where the decision to provide lighting has been taken by any authority working within national legislation or other constraints. The report is based on photometric considerations, and all values of luminance and illuminance are maintained values. The main body of the report covers the common aspects of Tunnel Lighting, and the various methods currently in use in Europe are detailed in the annexes. No single method is recommended.
NS-EN 13201-3:2003/A	2012-10-15	Norw	Corrigendum AC - Road lighting - Part 3: Calculation of performance	
CEN/TR 13201-1:2004	2004-07-01	Norw	Road lighting - Part 1: Selection of lighting classes	
NS-EN 13201-4:2003	2004-01-12	Norw	Road lighting - Part 4: Methods of measuring lighting performance	
NS-EN 13201-3:2003	2004-01-12	Norw	Road lighting - Part 3: Calculation of performance - (Corrigendum AC:2005 incorporated)	
NS-EN 13201-2:2003	2004-01-12	Norw	Road lighting - Part 2: Performance requirements	
PKN-CEN/TR 13201-1:2	2007-01-12	PKN (PL)	Road lighting -- Part 1: Selection of lighting classes	
PN-EN 13201-2:2007 P	2007-08-28	PKN (PL)	Road lighting - Part 2: Performance requirements	Defines, according to photometric requirements, lighting classes for road lighting aiming at the visual needs of road users, and it considers environmental aspects of road lighting
PN-EN 13201-3:2007 P	2007-10-11	PKN (PL)	Road lighting - Part 3: Calculation of performance	Describes the calculation methods, which enable road lighting quality characteristics to be calculated by agreed procedures so that results obtained from different sources will have a uniform basis
PN-EN 13201-4:2007 P	2007-09-13	PKN (PL)	Road lighting - Part 4: Methods of measuring lighting performance	Establishes conventions and procedures for lighting measurements of road lighting installations and gives advice on the use and selection of luminance meters and illuminance meters. Conditions which may lead to inaccuracies are identified and precautions are given to minimize these. A format for the presentation of the measurements is suggested
SS-EN 13201-3/AC:200	2007-03-05	SIS Swe	Road lighting - Part 3: Calculation of performance	
SS-EN 13201-4	2003-12-12	SIS Swe	Road lighting - Part 4: Methods for measuring lighting performance	
SS-EN 13201-3	2003-12-12	SIS Swe	Road lighting - Part 3: Calculation of performance	
SS-EN 13201-2	2003-12-12	SIS Swe	Road lighting - Part 2: Performance requirements	
SS-EN 40-7	2003-01-17	SIS Swe	Lighting columns - Part 7: Requirements for fibre reinforced polymer composite lighting columns	
SN EN 40-2	2005-01-00	SNV Swiss	Lighting columns - Part 2: General requirements and dimensions	
SN EN 40-3-1	2013-04-00	SNV Swiss	Lighting columns - Part 3-1: Design and verification - Specification for characteristic loads	
SN EN 40-3-2	2013-04-00	SNV Swiss	Lighting columns - Part 3-2: Design and verification - Verification by testing	
SN EN 40-4	2006-03-00	SNV Swiss	Lighting columns - Part 4: Requirements for reinforced and prestressed concrete lighting columns	
SN EN 40-4/AC	2012-05-00	SNV Swiss	Lighting columns - Part 4: Requirements for reinforced and prestressed concrete lighting columns; Corrigendum AC	
SN EN 40-5	2002-04-00	SNV Swiss	Lighting columns - Part 5: Requirements for steel lighting columns	
SN EN 40-6	2002-04-00	SNV Swiss	Lighting columns - Part 6: Requirements for aluminium lighting columns	
SN EN 40-7	2003-01-00	SNV Swiss	Lighting columns - Part 7: Requirements for fibre reinforced polymer composite lighting columns	
SN EN 13201-2	2004-03-00	SNV Swiss	Road lighting - Part 2: Performance requirements	
SN EN 13201-3	2004-01-00	SNV Swiss	Road lighting - Part 3: Calculation of performance	
SN EN 13201-3/AC	2007-05-00	SNV Swiss	Road lighting - Part 3: Calculation of performance; Corrigendum AC	
SN EN 13201-4	2004-03-00	SNV Swiss	Road lighting - Part 4: Methods of measuring lighting performance	
UNI 11248:2012	2012-10-04	UNI - It	Road lighting - Selection of lighting classes	This standard determines the lighting classes necessary to apply the performance requirements of UNI EN 13201-2 for assiuring lighting conditions suited for traffic safety and gives recomandations to obtain enery saving and to optimizing lighting plants and as regards to reduction of enviromental impact.
UNI 11431:2011	2011-11-10	UNI - It	Light and lighting - Use of luminous flux controllers in road lighting	This standard gives information for using luminous flux controllers in road lighting systems when the UNI EN 13201- 2 lighting classes are adopted and it describes methods to verify the energy saving their use can produce. The standard defines:- parameters to characterize limits and operating conditions of luminaires, when supplied with a luminous flux controller;- performance parameters of luminous flux controllers useful to define their conditions in road lighting applications;- typical control cycles to evaluate, during the design of a lighting system, the obtainable energy saving if a luminous flux controller is installed.
EC 1-2007 UNI EN 132(	2007-11-30	UNI - It	Road lighting - Part 3: Calculation of performance	Errata Corrige 1 of 30-11-2007 to UNI EN 13201-3:2004
UNI EN 40-7:2008	2008-01-24	UNI - It	Lighting columns - Parte 7: Requirements for fibre reinforced polymer composite lighting columns	This part of this European Standard specifies the performance requirements for fibre reinforced polymer composite lighting columns for which the main intended use is road lighting. It includes materials and test methods. The composite materials considered are those constructed from a fibrous reinforcing material that is suspended in a matrix of resin material. It applies to post top columns not exceeding 20 m height for post top.
UNI EN 13201-4:2004	2004-09-01	UNI - It	Road lighting - Part 4: Methods of measuring lighting performance	
UNI EN 13201-3:2004	2004-09-01	UNI - It	Road lighting - Part 3: Calculation of performance	
UNI EN 13201-2:2004	2004-09-01	UNI - It	Road lighting - Part 2: Performance requirements	
EC 1-2013 UNI 11248:2	2013-12-10	UNI - It	Road lighting - Selection of lighting classes	Errata corrige 1 of 10/12/2013 to UNI 11248:2012
EC 1-2012 UNI 11431:2	2012-06-14	UNI - It	Light and lighting - Use of luminous flux controllers in road lighting	Errata corrige 1 of 14-06-2012 to UNI 11431:2011
STN EN 40-2	2005-09-01	UNMS (SL)	Lighting columns. Part 2: General requirements and dimensions	
STN EN 40-4	2006-07-01	UNMS (SL)	Lighting columns. Part 4: Requirements for reinforced and prestressed concrete lighting columns	
STN EN 40-5	2002-12-01	UNMS (SL)	Lighting columns. Part 5: Requirements for steel lighting columns	
STN EN 40-6	2002-12-01	UNMS (SL)	Lighting columns. Part 6: Requirements for aluminium lighting columns	
STN EN 40-7	2003-08-01	UNMS (SL)	Lighting columns - Part 7: Requirements for fibre reinforced polymer composite lighting columns	
TNI CEN/TR 13201-1	2005-03-01	UNMS (SL)	Road lighting. Part 1: Selection of lighting classes	
STN EN 13201-2	2005-01-01	UNMS (SL)	Road lighting. Part 2: Performance requirements	
STN EN 13201-3	2005-01-01	UNMS (SL)	Road lighting. Part 3: Calculation of performance	
STN EN 13201-4	2005-01-01	UNMS (SL)	Road lighting. Part 4: Methods of measuring lighting performance	
NF C17-200	2007-03-01	UTE	External lighting installations - Requirements	
C17-201	1997-05-01	UTE	Public lighting installations. Rules.	
UTE C17-202U*UTE C1	2007-03-01	UTE	Installations of temporary illumination by garlands, lighting decorative systems or luminaries	
AFNOR C17-205	2014-08-01	UTE	Outdoor lighting installation - Determination of cross sectional Area of conductors and selection of protective devices	
UTE C17-210U*UTE C1	2003-08-01	UTE	Public lighting installations - Automatic disconnecting devices for public lighting	
UTE C17-260/A1U*UTE	2011-11-01	UTE	Outdoor lighting installation - Maintenance	
UTE C17-260U*UTE C1	2010-10-01	UTE	Outdoor lighting installation - Maintenance	
NF C71-003*NF EN 605	2003-06-01	UTE	Luminaires - Part 2-3 : particular requirements - Luminaires for road and street lighting	
NF C71-003/A1*NF EN	2011-11-01	UTE	Luminaires - Part 2-3 : particular requirements - Luminaires for road and street lighting	